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Uncertainty Estimation in Medical Imaging Segmentation with Deep Learning

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Abstract

In recent years, deep learning has demonstrated remarkable performance in medical image segmentation tasks. However, most state-of-the-art models provide deterministic predictions without accounting for uncertainty, which is a critical aspect in clinical decision-making. This thesis focuses on the estimation and visualization of uncertainty in medical image segmentation using neural networks. The work is based on the nnUNet framework, a robust and self-configuring pipeline for medical segmentation, which has been extended with probabilistic methods to quantify both aleatoric and epistemic uncertainty.

The experimental evaluation was conducted on a private dataset of 45 full-body CT scans. Several uncertainty estimation and calibration approaches were implemented, including Temperature Scaling, Deep Ensembles, Checkpoint Ensembles, and Test-Time Augmentation. Each method was integrated into the nnUNet v2 pipeline and evaluated using standard segmentation metrics (such as Dice Similarity Coefficient, DSC) and uncertainty-specific metrics (such as Expected Calibration Error, ECE).

The results show that models calibrated with temperature scaling ($T = 3.0$) achieve a significant reduction in ECE from 5.45% to 2.39%, without compromising segmentation quality (average DSC remained above 0.81). Deep Ensembles and Checkpoint Ensembles further improved uncertainty estimation, capturing epistemic variability across folds or training snapshots. Moreover, Test-Time Augmentation proved effective in identifying aleatoric uncertainty, highlighting ambiguous regions caused by poor contrast or anatomical complexity.

A key contribution of this work is the development of advanced entropy-based uncertainty visualizations, which were integrated into an interactive tool. These visual cues enable clinicians to inspect segmentation predictions alongside their confidence maps, revealing potentially unreliable regions in real time.

In conclusion, this thesis demonstrates that uncertainty-aware models can provide more trustworthy and interpretable predictions while maintaining high segmentation performance. By combining multiple estimation strategies with intuitive visualization tools, the proposed pipeline contributes to bridging the gap between AI models and clinical

adoption.

Keywords: medical image segmentation, uncertainty estimation, deep learning, nnUNet, temperature scaling, deep ensembles, checkpoint ensembles, test-time augmentation, entropy visualization, CT

Abstract in lingua italiana

Negli ultimi anni, il deep learning ha dimostrato prestazioni straordinarie nei compiti di segmentazione di immagini mediche. Tuttavia, la maggior parte dei modelli allo stato dell'arte produce predizioni deterministiche, senza tenere conto dell'incertezza, un aspetto cruciale nel processo decisionale clinico. Questa tesi si concentra sulla stima e visualizzazione dell'incertezza nella segmentazione di immagini mediche mediante reti neurali. Il lavoro si basa sul framework nnUNet, una pipeline robusta e auto-configurante per la segmentazione medica, estesa con metodi probabilistici per quantificare sia l'incertezza aleatoria che quella epistemica.

La valutazione sperimentale è stata condotta su un dataset privato costituito da 45 scansioni TC del corpo intero. Sono stati implementati diversi approcci per la stima dell'incertezza e la calibrazione, tra cui Temperature Scaling, Deep Ensembles, Checkpoint Ensembles e Test-Time Augmentation. Ogni metodo è stato integrato nella pipeline di nnUNet v2 ed è stato valutato sia tramite metriche standard di segmentazione (come il Dice Similarity Coefficient, DSC) sia con metriche specifiche per l'incertezza (come l'Expected Calibration Error, ECE).

I risultati mostrano che i modelli calibrati con temperature scaling ($T = 3.0$) ottengono una riduzione significativa dell'ECE, passando dal 5.45% al 2.39%, senza compromettere la qualità della segmentazione (il DSC medio è rimasto superiore a 0.81). I Deep Ensemble e i Checkpoint Ensemble hanno ulteriormente migliorato la stima dell'incertezza, catturando la variabilità epistemica tra fold o durante le fasi di training. Inoltre, il Test-Time Augmentation si è dimostrato efficace nell'identificare l'incertezza aleatoria, evidenziando regioni ambigue dovute a scarso contrasto o complessità anatomica.

Un contributo chiave di questo lavoro è lo sviluppo di visualizzazioni avanzate dell'incertezza basate sull'entropia, integrate in uno strumento interattivo. Queste rappresentazioni grafiche permettono ai medici di ispezionare le predizioni del modello insieme alle mappe di confidenza, rivelando in tempo reale le aree potenzialmente inaffidabili.

In conclusione, questa tesi dimostra che modelli consapevoli dell'incertezza possono offrire predizioni più affidabili e interpretabili, mantenendo al contempo elevate prestazioni di

segmentazione. Combinando diverse strategie di stima con strumenti di visualizzazione intuitivi, la pipeline proposta contribuisce a colmare il divario tra i modelli di intelligenza artificiale e la loro effettiva adozione clinica.

Parole chiave: segmentazione di immagini mediche, stima dell'incertezza, deep learning, nnUNet, temperature scaling, deep ensembles, checkpoint ensembles, test-time augmentation, visualizzazione dell'incertezza, tomografia computerizzata

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1 | Introduction

Deep learning has revolutionized medical image analysis, enabling the development of powerful models capable of performing complex tasks such as detection, classification, and segmentation with unprecedented levels of accuracy. Among these, *semantic segmentation*—the process of assigning a class label to each pixel or voxel in an image—has emerged as a key component in computer-assisted diagnosis, radiotherapy planning, and anatomical structure delineation.

However, despite their remarkable performance, deep learning models remain largely opaque in terms of interpretability. Their predictions are often treated as deterministic, with little or no indication of confidence or reliability. In the clinical domain, where decisions based on automated analysis can have life-altering consequences, this opacity raises serious concerns. A model that appears highly confident about a wrong prediction may lead to diagnostic errors or missed treatment opportunities. As such, the ability to estimate and communicate uncertainty is increasingly recognized as a critical requirement for the safe and trustworthy deployment of AI in healthcare.

This issue is particularly relevant in the context of **medical image segmentation**, where complex anatomical variability, low-contrast imaging, motion artifacts, and inter-observer disagreement contribute to significant ambiguity. In such scenarios, knowing where and to what extent a model is uncertain can provide valuable insights for clinicians, allowing them to focus attention on problematic regions, verify predictions, or prioritize manual correction. The incorporation of uncertainty estimation into segmentation pipelines is therefore a promising direction toward building more transparent and clinically useful AI systems.

Motivation and Clinical Context

This thesis focuses on a specific clinical application: the segmentation of the **lymphatic system** from full-body computed tomography scans (CT). The lymphatic system plays a central role in immune function and cancer staging, particularly in conditions such as

lymphoma or metastatic progression. However, its segmentation from CT is notoriously difficult: lymph nodes are small, poorly contrasted, and often indistinguishable from surrounding tissues in standard imaging protocols. Manual delineation is time-consuming, labor-intensive, and prone to high inter-observer variability.

To address this challenge, the study leverages **nnU-Net v2**, a state-of-the-art deep learning framework that automatically configures its architecture and training pipeline based on the dataset characteristics. While **nnU-Net** provides a strong segmentation baseline, its outputs—like those of most deep learning models—do not include uncertainty estimates by default. The core motivation of this work is therefore to *extend a powerful segmentation framework with robust, interpretable, and clinically meaningful uncertainty estimation tools*.

Research Objectives

The overarching goal of this thesis is to investigate, implement, and evaluate techniques for uncertainty estimation in deep learning-based medical image segmentation. To this end, the work first explores the theoretical underpinnings of uncertainty in machine learning, with particular emphasis on the distinction between epistemic uncertainty—stemming from model limitations or insufficient data—and aleatoric uncertainty, which arises from intrinsic noise in the input data.

Building on this foundation, several estimation techniques are implemented within the **nnU-Net v2** pipeline. These include post-hoc calibration through temperature scaling, the use of deep ensembles to capture variability across independently trained models, test-time augmentation to probe input-dependent uncertainty, and checkpoint-based posterior sampling to approximate Bayesian inference over training dynamics. Each method is adapted and integrated into a unified experimental framework designed to assess their practical value in a clinically relevant setting.

In parallel, entropy-based visualization strategies are developed to transform the probabilistic outputs of these models into interpretable heatmaps of voxel-wise uncertainty. These visual tools are essential for translating raw confidence scores into insights that clinicians can interpret in real time. The models are evaluated not only in terms of their segmentation accuracy—measured via Dice Similarity Coefficient—but also based on the quality of their uncertainty estimates, using metrics such as Expected Calibration Error (ECE) and Brier Score.

Beyond global performance, the spatial correlation between regions of high uncertainty

and actual prediction errors is systematically analyzed. This analysis also includes the exploration of thresholding strategies to isolate areas of potential clinical interest, particularly in anatomically ambiguous or error-prone regions.

To assess the real-world applicability of these methods, a second evaluation phase is conducted on a set of segmentations manually corrected by expert physicians. This phase provides a more realistic benchmark for the clinical utility of uncertainty maps and supports the hypothesis that uncertainty-aware systems can aid in identifying and correcting model failures.

Finally, an interactive application is developed to visualize and explore both segmentation results and their associated uncertainty maps. This tool lays the groundwork for future integration into human-in-the-loop clinical workflows, where automated predictions are supplemented by interpretable cues and expert feedback.

Main Contributions and Results

The experiments conducted in this thesis demonstrate that integrating uncertainty estimation into a segmentation pipeline can significantly improve both the interpretability and the trustworthiness of model predictions. Among the various configurations tested, the combination of ensemble inference, moderate test-time augmentation, and temperature scaling with $T = 3$ emerged as the best-performing strategy, achieving a Dice score of 0.8145 and the lowest Expected Calibration Error (ECE) and Brier Score across the test set.

Entropy-based uncertainty maps proved to be highly effective in highlighting ambiguous regions, particularly those corresponding to actual segmentation errors. These maps were further refined through a principled thresholding strategy and successfully validated in a second evaluation phase involving 15 additional patients. In this phase, the high-entropy areas consistently aligned with the regions corrected by expert physicians, confirming the clinical relevance of the proposed methods.

To enhance usability and support clinical integration, an interactive application was developed, allowing real-time visualization of segmentation masks, uncertainty heatmaps, and ground truth overlays. This tool facilitates slice-by-slice navigation across anatomical planes and supports customizable filtering of uncertainty based on entropy thresholds, offering an intuitive interface for both researchers and clinicians.

Structure of the Thesis

The thesis is structured as follows:

- **Chapter 2** introduces the theoretical foundations of this work, including the fundamentals of image segmentation, the U-Net architecture, and the concepts of predictive uncertainty and model calibration in deep learning.
- **Chapter 3** reviews the state of the art in uncertainty estimation for medical image segmentation. It presents the most relevant methods from the literature and outlines their motivations, methodological differences, and clinical implications.
- **Chapter 4** describes the dataset used in the experiments and details the training configurations, preprocessing steps, and uncertainty estimation techniques implemented within the **nnU-Net v2** framework. It also discusses the design and generation of entropy-based uncertainty visualizations.
- **Chapter 5** presents the results of both quantitative and qualitative evaluations, including segmentation accuracy, calibration quality, and entropy analysis. It also introduces the second evaluation phase based on expert-corrected masks and showcases the functionality of the developed Streamlit application for interactive visualization.
- **Chapter 6** concludes the thesis by summarizing the main findings, discussing the limitations of the current study, and suggesting possible directions for future research.

2 | Background

2.1. Overview

This chapter provides the theoretical foundation necessary to understand the methods and contributions presented in the following chapters. We begin by introducing the task of medical image segmentation, highlighting its challenges and clinical relevance. We then present the U-Net architecture, which has become a *de facto* standard in biomedical segmentation tasks due to its effectiveness and versatility.

In the second part of the chapter, we turn to the concept of uncertainty in machine learning. We define and differentiate between epistemic and aleatoric uncertainty, and examine their formalization within a Bayesian framework. These notions are central to the development of reliable AI systems, particularly in safety-critical domains such as medical imaging. Finally, we discuss the concept of model calibration and its importance in interpreting neural network outputs as meaningful confidence estimates.

These foundational concepts set the stage for the practical implementation of uncertainty-aware segmentation methods presented in later chapters.

2.2. Medical Image Segmentation

Medical image segmentation plays a central role in medical image analysis, acting as a critical prerequisite for accurate diagnosis and treatment planning. It refers to the process of delineating anatomical structures or pathological regions (such as organs or lesions) from modalities like CT, MRI, or X-ray images. Traditional segmentation approaches include thresholding, edge detection, and morphological operations, which although interpretable, often suffer from inefficiency, low scalability, and strong dependence on expert-designed features. These handcrafted methods are generally inadequate for large-scale, diverse, and high-resolution medical imaging data.

To address these limitations, deep learning-based models have become the prevailing solution in recent years. Convolutional Neural Networks (CNNs), particularly architectures designed for semantic segmentation, have been extensively employed due to their capability to learn hierarchical features automatically. Among these, the **U-Net** model has become the *de facto* standard for medical image segmentation. Its encoder-decoder structure with skip connections enables it to preserve spatial resolution while extracting rich semantic information [35].

2.2.1. U-Net Architecture for Biomedical Image Segmentation

One of the most influential architectures in the field of biomedical image segmentation is the U-Net, proposed by Ronneberger et al. in 2015 [23]. This network was specifically designed to address the challenges commonly encountered in medical imaging tasks, such as the scarcity of annotated data and the need for high spatial precision. Unlike traditional classification-focused convolutional neural networks (CNNs), U-Net is tailored for pixel-wise prediction tasks, where each pixel must be assigned a class label corresponding to a specific anatomical or pathological structure.

Architecture Overview. The U-Net follows a symmetrical encoder-decoder architecture composed of two main paths: the contracting path (encoder) and the expansive path (decoder). The contracting path is analogous to a typical CNN and is responsible for capturing the contextual features of the input image. It comprises repeated applications of two 3×3 convolutions (without padding), each followed by a ReLU activation and a 2×2 max pooling operation with stride 2 for downsampling. With each downsampling step, the number of feature channels is doubled, allowing the network to capture increasingly abstract and hierarchical representations (see Figure 2.1).

The expansive path, in contrast, aims to restore spatial resolution and enable precise

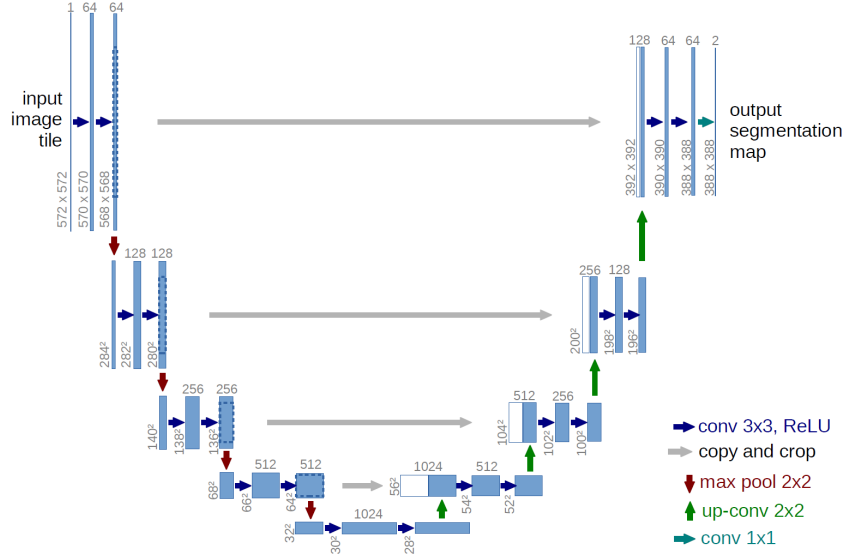


Figure 2.1: U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations [33].

localization. It consists of upsampling operations followed by 2×2 “up-convolutions” (transposed convolutions) that halve the number of feature channels, concatenation with the corresponding cropped feature maps from the contracting path (via skip connections), and two 3×3 convolutions followed by ReLU activations. The skip connections allow the network to propagate high-resolution spatial information from the encoder to the decoder, enhancing the localization capabilities of the model. The final layer is a 1×1 convolution that maps each 64-component feature vector to the desired number of segmentation classes.

Training Strategy and Loss Function. To accommodate the lack of large annotated datasets in the biomedical domain, U-Net employs an extensive data augmentation scheme, including random elastic deformations, which are particularly effective in simulating realistic anatomical variability. The network is trained end-to-end using stochastic gradient descent (SGD) with a cross-entropy loss applied pixel-wise. To further improve performance in segmenting touching or overlapping objects (e.g., adjacent cells), the authors introduced a class balancing and boundary-aware loss. This is implemented via a precomputed pixel-wise weight map, which assigns higher weights to pixels on borders between objects, effectively forcing the network to focus on these difficult regions.

Formally, the loss function is defined as:

$$E = \sum_{x \in \Omega} w(x) \cdot \log(p_{\ell(x)}(x)),$$

where $w(x)$ is the weight assigned to pixel x , and $p_{\ell(x)}(x)$ is the predicted softmax probability for the true class $\ell(x)$. The weight map $w(x)$ includes a Gaussian term centered on the borders between objects to enforce separation.

2.3. nnU-Net: A Self-Adapting Framework for Medical Image Segmentation

The emergence of U-Net has revolutionized medical image segmentation, but applying it effectively to new datasets often required significant manual effort. For each specific task, researchers had to carefully adapt the network architecture, preprocessing pipeline, training regimen, and postprocessing heuristics, leading to a proliferation of custom solutions that hindered reproducibility and generalization. This context created the ideal ground for the development of *nnU-Net* — the “no-new-Net” framework introduced by Isensee et al. [13] — which aims not to reinvent network architectures, but to standardize and automate the configuration process of medical image segmentation pipelines.

At its core, nnU-Net is not just a single model or architecture, but a fully automated framework capable of adapting itself to the idiosyncrasies of each new dataset. It operationalizes a set of best practices derived from empirical evidence across numerous challenges and benchmarks, allowing it to produce strong baseline results without any manual tuning. This philosophy turns nnU-Net into a kind of “meta-learner” for segmentation pipelines: instead of designing a new model from scratch for each problem, nnU-Net intelligently configures U-Net-based architectures by analyzing dataset characteristics and selecting appropriate design choices.

What distinguishes nnU-Net from traditional approaches is its ability to handle this configuration in a comprehensive and hierarchical manner. The framework first analyzes the dataset to infer properties such as image spacing, modality, class imbalance, and image shape. Based on this information, it selects one or more of three architectural variants: a 2D U-Net for highly anisotropic data (such as thick-slice MR scans), a full-resolution 3D U-Net for smaller isotropic volumes, and a cascaded 3D U-Net for larger volumes that require a coarse-to-fine approach. The cascaded setup, in particular, is a clever two-stage design where the first network predicts a rough segmentation at low resolution,

which is then refined by a second network working at full resolution and informed by the preliminary result.

The architectural design itself remains close to the original U-Net. It employs multiple resolution levels, with two convolutional blocks per level, instance normalization, and leaky ReLU activation. Rather than fixing the depth and width of the model, nnU-Net adapts these parameters dynamically to respect GPU memory constraints, striking a balance between computational feasibility and representational capacity.

Beyond architecture, nnU-Net exerts a powerful influence over all stages of the segmentation pipeline. During preprocessing, it normalizes images in modality-specific ways (global normalization for CT, per-case for MR), resamples them to a common voxel spacing, and crops non-informative background to focus on regions of interest. The training phase is designed to be robust and reproducible, combining cross-entropy and Dice losses with extensive data augmentation techniques, including affine and elastic deformations, gamma transformations, and mirroring. Optimization is performed using the Adam algorithm with adaptive learning rate schedules, and validation loss is used to trigger early stopping. At inference time, nnU-Net applies sliding window predictions augmented with test-time transformations and optionally ensembles results from multiple cross-validation folds or architectural variants.

One of the strengths of nnU-Net lies in its parameter configuration strategy, which distinguishes between three types of parameters. Some are globally fixed based on prior knowledge of best practices; others are derived using heuristic rules based on dataset properties; and a third group is determined empirically through internal validation. This structured approach ensures that the pipeline is not only adaptable but also systematically optimized for performance.

The results speak for themselves. In the Medical Segmentation Decathlon (MSD), a benchmark involving ten diverse segmentation tasks, nnU-Net achieved top-ranking performance in seven tasks and delivered the best overall score. Its success was not limited to the MSD: it also excelled in BraTS, KiTS, ACDC, BTCV, and numerous other challenges. Importantly, the open-source release of nnU-Net catalyzed widespread adoption, rapidly becoming a de facto standard in the field. For instance, in MICCAI competitions following its publication, a large majority of winning teams reported using nnU-Net or a close derivative, attesting to its robustness and utility.

In conclusion, nnU-Net represents a paradigm shift in the way we approach medical image segmentation. By abstracting away the need for manual engineering and embedding strong empirical priors into an automated pipeline, it democratizes access to high-quality

segmentation models. Its architecture-agnostic, data-driven, and self-configuring nature has had a profound impact on the field, making it an essential tool in both academic research and translational clinical applications.

2.4. TotalSegmentator: Automatic Full-Body Segmentation in CT and MRI Imaging

One of the most impactful recent developments in medical image segmentation is the introduction of the open-source framework **TotalSegmentator**, designed to automatically segment over 100 anatomical structures from volumetric medical images—most notably computed tomography (CT) and magnetic resonance imaging (MRI). Originally proposed by Wasserthal et al. in 2023 [31], TotalSegmentator stands out for its ability to perform large-scale, full-body, multi-structure segmentation with expert-level accuracy, while remaining user-friendly, highly generalizable, and easily integrable into both research and clinical workflows [2].

The motivation behind TotalSegmentator was to overcome three major limitations of prior segmentation tools: the fragmentation of available models (typically trained for a single organ or anatomical region), their poor generalizability to real-world data outside controlled benchmarks, and their technical inaccessibility for non-expert users. To address these issues, TotalSegmentator builds upon the **nnU-Net** framework [13], which provides a self-configuring pipeline capable of adapting its architecture and training strategy to the properties of the given dataset. Leveraging this capability, TotalSegmentator offers a unified, general-purpose segmentation system with minimal manual intervention.

The underlying architecture is based on a three-dimensional U-Net, following a symmetric encoder-decoder configuration. One of its defining features is a *modular design*: while the main “total” model segments over 100 structures in a single pass, additional task-specific variants exist—such as those for craniofacial, brain, or thoracic regions—trained on dedicated datasets and callable through a unified API or command-line interface [32].

The CT model was trained on a heterogeneous dataset comprising 1,204 whole-body CT scans from patients with varying ages, pathologies, acquisition protocols, and scanner types. Each image was annotated for 104 anatomical structures, including major parenchymal organs (e.g., liver, spleen, kidneys, pancreas), vascular structures (e.g., aorta, vena cava, iliac arteries), skeletal elements (vertebrae and ribs indexed individually), and muscle groups. Annotation was performed semi-automatically using an iterative loop where preliminary models aided human annotators, drastically reducing the time required

without sacrificing label quality [31].

The performance of the CT model is remarkable. On its internal test set, TotalSegmentator achieved a **mean Dice score of 0.943** across all structures, indicating a very high degree of overlap between predicted and expert-generated masks. The model maintained strong accuracy even in cases of noisy or pathological scans. External validation further demonstrated its robustness: when evaluated on a separate dataset of 4,004 whole-body CT scans, TotalSegmentator significantly outperformed another open-source multi-organ segmentation model (Dice 0.932 vs 0.871, $p < 0.001$) [31].

Building upon the success of the CT version, a new model—**TotalSegmentator MRI**—was developed for use with MRI data. This model was trained on 616 full-body MRI studies, augmented with more than 500 CT scans to improve multi-modal generalization. Notably, the MRI model is *sequence-agnostic*: it performs robustly across T1- and T2-weighted sequences, with or without contrast enhancement, despite being trained on highly heterogeneous data [2]. The MRI model covers approximately 80 anatomical structures and achieved a **mean Dice score of around 0.84**, with values exceeding 0.9 for well-contrasted organs such as the kidneys and liver. Public benchmark evaluations—such as those on the CHAOS and AMOS datasets—confirmed its superiority over competing models, showing significant performance gains on both overlap metrics and generalization ability.

Beyond segmentation accuracy, TotalSegmentator is also notable for its practical usability. It is distributed via PyPI as a lightweight Python package, compatible with modern GPUs, and includes a dedicated extension for 3D Slicer to enable immediate deployment in clinical environments. This accessibility opens up numerous real-world applications: organ volumetry for oncological planning, body composition analysis for metabolic studies, automatic delineation of organs-at-risk in radiation therapy, and opportunistic screening (e.g., measuring bone density or visceral fat from existing scans) are all made feasible without manual intervention [2, 31].

Moreover, TotalSegmentator represents a cultural shift in the medical AI community. Unlike many academic models that remain unpublished or inaccessible, TotalSegmentator is fully open-source and well-documented, encouraging reproducibility and collaborative improvement. Its widespread adoption—reaching hundreds of thousands of users within two years—demonstrates the potential of combining deep learning generalists, auto-configuring frameworks, and clinically representative datasets to bridge the gap between experimental research and real-world impact.

In conclusion, TotalSegmentator represents a significant advance in medical image seg-

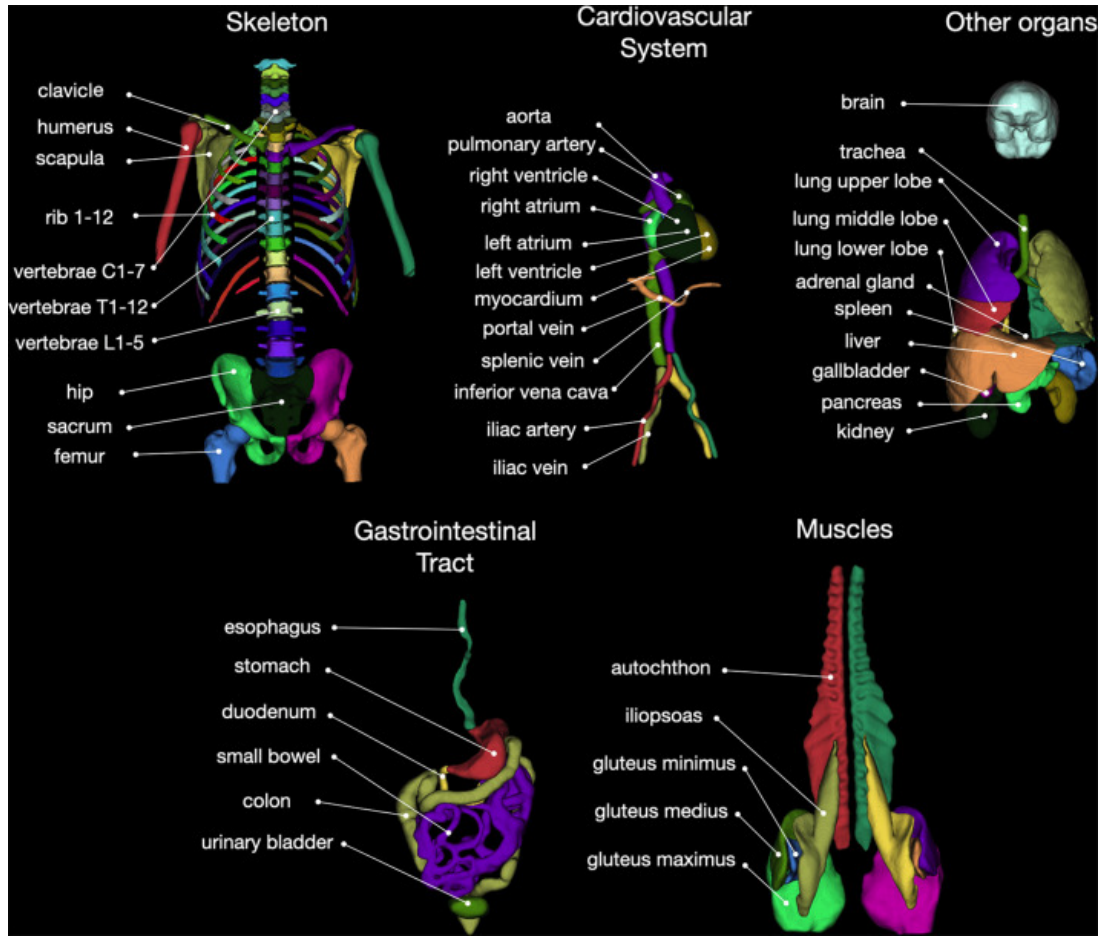


Figure 2.2: Example of multi-structure segmentation output from TotalSegmentator on a full-body CT scan. Each color corresponds to a different anatomical region. Source: [31].

mentation, offering high accuracy, broad anatomical coverage, and seamless usability. Its architecture, grounded in nnU-Net, paired with large-scale curated training data and robust engineering design, positions it as a state-of-the-art tool in both research and clinical applications.

2.5. Uncertainty

Uncertainty is a topic that pervades many areas of daily life—for example, in weather forecasts, sports, and increasingly in healthcare, where emerging technologies assist doctors in diagnosing diseases. Achieving a reliable estimate of uncertainty, especially in healthcare, is therefore of critical importance.

When working with data and machine learning models, as mentioned in [29], we generally encounter two types of uncertainty: **Aleatoric Uncertainty** (also known as Data Uncertainty) and **Epistemic Uncertainty** (or Knowledge Uncertainty).

2.5.1. Aleatoric Uncertainty

Aleatoric Uncertainty primarily arises from the variability in data distribution, which makes it challenging, or sometimes even impossible, to classify data accurately due to the absence of distinguishing features.

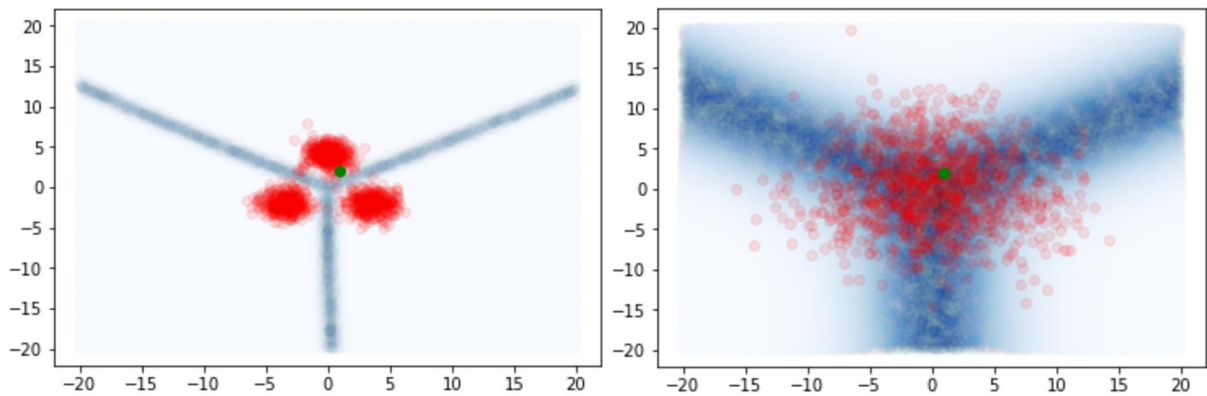


Figure 2.3: In the plot on the left, the green point can be classified without issues because there is no data uncertainty; we can clearly distinguish that it falls within a well-defined distribution that belongs to a specific class. The distribution is distinct, and there is a clear boundary separating it from others, making classification straightforward. The plot on the right, instead, illustrates data uncertainty, as there appears to be no discernible pattern to differentiate the classes, making it impossible to attribute the new data point (green) to any class with certainty. Here, the overlapping distributions make classification ambiguous, highlighting the presence of aleatoric (data-related) uncertainty.

In regression tasks, data uncertainty further divides into two types:

- **Homoscedastic:** Input-independent, where the noise in the data distribution is uniform and unrelated to the inputs.

- **Heteroscedastic:** Input-dependent, where the noise correlates with the underlying data distribution.

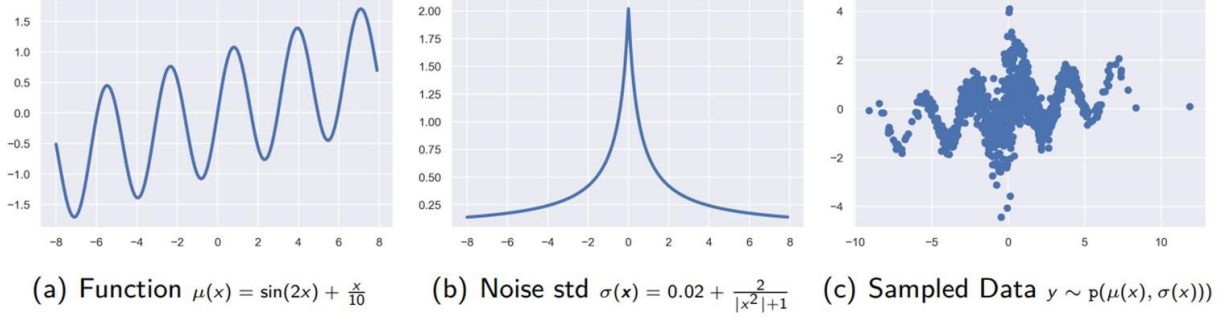


Figure 2.4: Example of regression with heteroscedastic uncertainty. (a) Mean function $\mu(x) = \sin(2x) + \frac{x}{10}$. (b) Noise standard deviation $\sigma(x) = 0.02 + \frac{2}{|x^2|+1}$ shows input dependency, with higher uncertainty near the origin. (c) Sampled data from $y \sim \mathcal{N}(\mu(x), \sigma(x))$, highlighting the noise variability.

Reasons for data uncertainty are varied, including measurement errors, data noise, sampling errors, data sparsity, ambiguity in data sources, biases in data collection, inconsistent data formats, or temporary uncertainty in observations.

2.5.2. Epistemic Uncertainty

Epistemic Uncertainty, on the other hand, refers to gaps in our understanding or knowledge about the model and data. This type of uncertainty can be subdivided into two main categories:

- **Data Sparsity:** This occurs in classification problems when one of the classes has very limited data (see figure 2.5). For instance, in a scenario with three classes, if one class has only a few samples, we experience significant knowledge uncertainty for this class, as we lack the knowledge to define it accurately.
- **Out-of-Distribution Data:** Here, the classes are well-balanced, and data is sufficiently abundant. However, a new data point might appear that is far from all classes (see figure 2.6), known as an “out-of-distribution” data point. This situation introduces knowledge uncertainty because the model or user lacks the knowledge to confidently assign this point to any existing class, as it lies outside the familiar distribution of the training data.

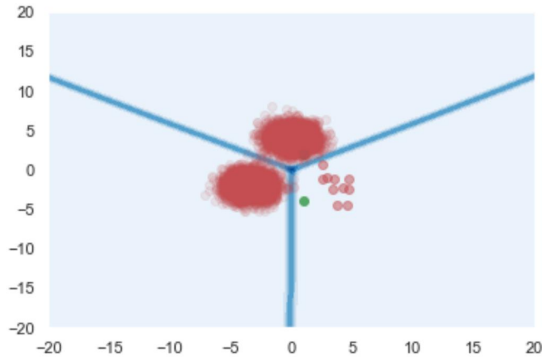


Figure 2.5: There is an high data sparsity in the class on the right-down region

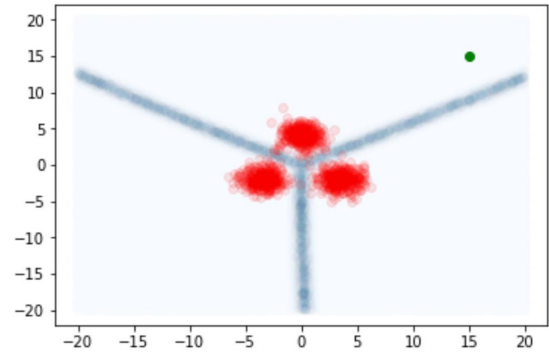


Figure 2.6: The new data point in green is out of distribution

2.5.3. Domain Shift

Knowledge uncertainty, encompassing both Data Sparsity and Out-of-Distribution data, can also be understood as “domain shift.” Domain shift is a concept often overlooked in deep learning applications, yet in real-world settings, it is almost unavoidable. When using data in real-world environments, the distribution of data may shift significantly from that observed during training—due to data sparsity or the presence of out-of-distribution data.

Let’s consider examples of domain shift.

In Figure 2.7, we see two MRI scans of the brain. These scans are often used to measure white matter lesions, which can serve as biomarkers for tracking the progression of multiple sclerosis. However, the two scans may have been produced using machines with different magnetic field strengths. For instance, the scan on the left might use a 7 Tesla magnet, while the one on the right uses a 1.5 Tesla field. In this example, it’s even visually clear that the images differ.

This highlights a common challenge: medical data might originate from hospitals that use specific equipment or parameters, such as particular scanners or magnetic field strengths. Training a model on such data may introduce knowledge uncertainty, as a domain shift could occur when different experimental settings or equipment are involved in data collection, leading to variations when applied to real-world scenarios with potentially different scanners.

Domain shift can arise in all forms of data, not just medical imaging.

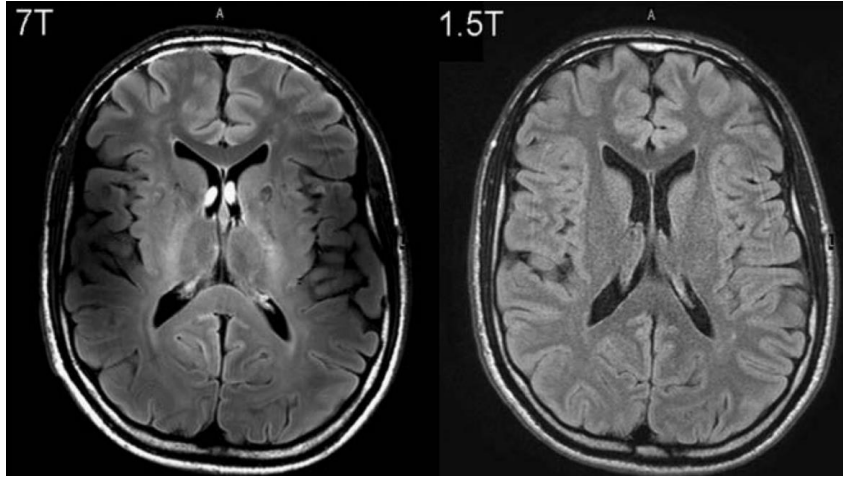


Figure 2.7: This image shows two MRI brain scans captured using different magnetic field strengths: 7 Tesla (left) and 1.5 Tesla (right). The difference in field strength affects image clarity and detail, demonstrating domain shift. Variations like these in medical imaging can lead to knowledge uncertainty when using AI models trained on data from only one type of scanner, as models may struggle with data from unfamiliar sources.

2.5.4. Bayesian Framework

Let's delve into the more mathematical side to truly understand what **data uncertainty** and **knowledge uncertainty** mean and how they contribute to the overall uncertainty in predicting any outcome, using the Bayesian framework.

Imagine we have a dataset, which we can call our training data, though we can label it in any way we prefer. This data consists of a set of inputs $\mathbf{X}$, such as images of cats, dogs, or medical scans. We have n such data points, $x_1, x_2, \dots, x_n$, which we will refer to collectively as $\mathbf{X}$. Each of these data points has a corresponding label, which could represent the class (e.g., cat or dog) or a numerical value (like 0, 1, 2, $\dots$). Thus, we have corresponding labels (or targets) $y_1, y_2, \dots, y_n$ for each input point, which we will refer to as $\mathbf{Y}$.

$$p(y^*|x^*, \mathbf{X}, \mathbf{Y}) = \int p(y^*|x^*, \omega) p(\omega|\mathbf{X}, \mathbf{Y}) d\omega \quad (2.1)$$

Using the Bayesian framework formally described in the equation 2.1, the objective is to provide the system with a new data point, called x^* , and obtain a prediction y^* . Essentially, we want to know the most probable output y^* given the input x^* and the type of training data that the model has seen. In the Bayesian approach, rather than aiming for a point prediction, we seek a **distribution** for our output instead of a single

value.

Referring to the equation, our goal is to derive the probability distribution on the left side of the equation, representing the probability distribution of y^* , given our input x^* and all the training data $\mathbf{X}$ and $\mathbf{Y}$. This distribution tells us, in a "soft" way, which values y^* are more likely, given our training data and the new input. Rather than providing a "hard" single prediction, this probabilistic approach gives us a measure of how confident we are in different possible predictions for y^* .

Ideally, if this distribution is very **sharp** (peaked), it means our uncertainty is low, as we are confident about the prediction. Conversely, if the distribution is very **flat**, it suggests a high degree of uncertainty, meaning y^* could potentially take on a wide range of values, with no single value standing out as particularly likely.

This is the distribution we are interested in. Typically, in the literature, to capture this distribution, we consider it as the integral of the likelihood function multiplied by the posterior distribution. But what does this mean?

In the equation, ω represents the **weights** that the model tries to learn during training. In typical deep learning architectures, these weights are hard values that we aim to learn. However, in the Bayesian perspective, we aim to learn a **distribution** for our weights. This is what we mean by the **posterior distribution**—the distribution over parameters that best describes the mapping between input x and output y . Therefore, the second term inside the integral in the equation represents our posterior distribution over the weights.

The first term, known as the **likelihood function**, represents the probability distribution for y^* given the input x^* and the weights, but it's not conditioned on the training data.

The final distribution we want to obtain, which is $p(y^*|x^*, \mathbf{X}, \mathbf{Y})$, is simply a **weighted average of our likelihood function**, with the posterior distribution providing the weights.

- The **likelihood function** encapsulates what we've defined as **data uncertainty**.
- The **posterior function uncertainty** refers to what we can call **knowledge uncertainty**.

This theoretical framework helps us understand where data uncertainty and knowledge uncertainty come from in a more formal way. However, these two types of uncertainty are typically **intractable**—in many real-world scenarios, we cannot obtain exact expressions for them. Therefore, what is often done is to approximate these distributions in various

ways to try to quantify these uncertainties.

2.5.5. Gaussian Processes

The Gaussian process is a powerful statistical tool that allows us to model distributions over functions. It has been widely employed in both supervised and unsupervised learning tasks, including regression and classification [22][28]. GPs provide advantageous properties such as uncertainty quantification for predictions, robustness to overfitting, and principled methods for hyperparameter optimization. Additionally, the use of approximate variational inference enables scaling to large datasets through stochastic and distributed inference [11], [9].

Consider a dataset consisting of N input-output pairs, $\{(x_1, y_1), \dots, (x_N, y_N)\}$, where the inputs $X \in \mathbb{R}^{N \times Q}$ and the outputs $Y \in \mathbb{R}^{N \times D}$. We aim to infer a function $y = f(x)$ that likely generated the observed data. To achieve this, a prior distribution is defined over the space of functions, $p(f)$, representing our belief about which functions are more or less likely to have generated the data. Using Bayes' theorem, we calculate the posterior distribution over the space of functions conditioned on the observed data as follows:

$$p(f | X, Y) \propto p(Y | X, f)p(f).$$

This posterior describes the functions that are most plausible given the observed dataset.

By employing Gaussian processes to model the function distribution, we can analytically evaluate the posterior for regression tasks and approximate it for classification tasks. Specifically, in regression, we assume a joint Gaussian distribution over the function values:

$$F | X \sim \mathcal{N}(0, K(X, X)), \quad Y | F \sim \mathcal{N}(F, \tau^{-1} I_N),$$

where $K(X, X)$ denotes the covariance matrix generated by the covariance function $K(x_i, x_j)$, τ is a precision hyperparameter, and I_N is the identity matrix of size N .

For classification, we use a categorical distribution obtained by applying a softmax function element-wise to Y :

$$F | X \sim \mathcal{N}(0, K(X, X)), \quad Y | F \sim \mathcal{N}(F, 0 \cdot I_N),$$

$$c_n | Y \sim \text{Categorical} \left(\frac{\exp(y_{n,d})}{\sum_{d'} \exp(y_{n,d'})} \right) \quad \text{for } n = 1, \dots, N.$$

Here, c_n is the observed class label for the n -th data point. Note that we treat Y and F differently for notational clarity, facilitating unified treatment of regression and classification.

To model the data, we must select a covariance function $K(x_i, x_j)$, which quantifies the similarity between pairs of input points. For a dataset of size N , this induces an $N \times N$ covariance matrix, $K(X, X)$. A common choice is the squared exponential covariance function, though non-stationary options such as those corresponding to Tanh (hyperbolic tangent) or ReLU (rectified linear unit) functions are also viable.

Computing the Gaussian distribution involves inverting the $N \times N$ covariance matrix, which has a computational cost of $\mathcal{O}(N^3)$. To handle larger datasets efficiently, various approximations to Gaussian processes have been developed.

2.6. Calibration of Predictive Confidence in Deep Learning

In the context of machine learning and, more specifically, deep neural networks, the concept of *calibration* refers to the alignment between a model’s predicted confidence and the true probability of a correct prediction. Although modern deep learning models are often highly accurate, they tend to be poorly calibrated, meaning that the confidence scores associated with their predictions are not reliable indicators of true correctness likelihood [10]. This misalignment can have critical implications in high-stakes fields such as healthcare, where downstream decisions may depend on the numerical values of predicted probabilities.

Calibration is intimately connected with uncertainty estimation. While uncertainty aims to express how unsure a model is about its predictions—either due to data variability (aleatoric) or lack of knowledge (epistemic)—calibration quantifies how much this uncertainty (or its complement, the confidence) can be trusted as a probabilistic estimate. In this sense, calibration does not create new uncertainty information but evaluates whether the model’s confidence truly corresponds to empirical reliability.

2.6.1. Understanding Calibration

To understand what it means for a model to be calibrated, consider a classifier that, when making predictions with 80% confidence, is indeed correct in 80% of those cases. Such a model is said to be *perfectly calibrated*. Conversely, a model that outputs 80% confidence but is only correct 60% of the time is considered *overconfident*.

Formally, given a classifier that outputs a confidence score $p(x)$ for an input x , the model is calibrated if:

$$\mathbb{P}(Y = \hat{Y} \mid p(x) = p) = p, \quad \forall p \in [0, 1] \quad (2.2)$$

where Y is the true label and $\hat{Y}$ is the predicted label. This definition extends naturally to segmentation tasks, where pixel-wise or voxel-wise confidence values can be interpreted in the same probabilistic framework.

In practice, most modern neural networks tend to be **overconfident**, especially when trained with the cross-entropy loss without any explicit regularization or uncertainty modeling.

2.6.2. Measuring Calibration

Several methods have been proposed to quantify the degree of (mis)calibration in neural networks. Among the most commonly used metrics are the following:

Expected Calibration Error (ECE). The ECE is a scalar summary of the difference between confidence and accuracy across all predictions. It is computed by dividing the predictions into M bins according to their confidence scores, and then calculating:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (2.3)$$

where B_m is the set of samples in bin m , $\text{acc}(B_m)$ is the average accuracy in that bin, $\text{conf}(B_m)$ is the average confidence, and n is the total number of samples. A well-calibrated model should yield $\text{acc}(B_m) \approx \text{conf}(B_m)$ in each bin.

Reliability Diagram. A graphical representation of calibration, the reliability diagram (see Figure 2.8) plots $\text{acc}(B_m)$ against $\text{conf}(B_m)$ for each bin. Perfect calibration corresponds to the diagonal line, where confidence matches empirical accuracy. Bars above the

diagonal indicate underconfidence, while bars below indicate overconfidence.

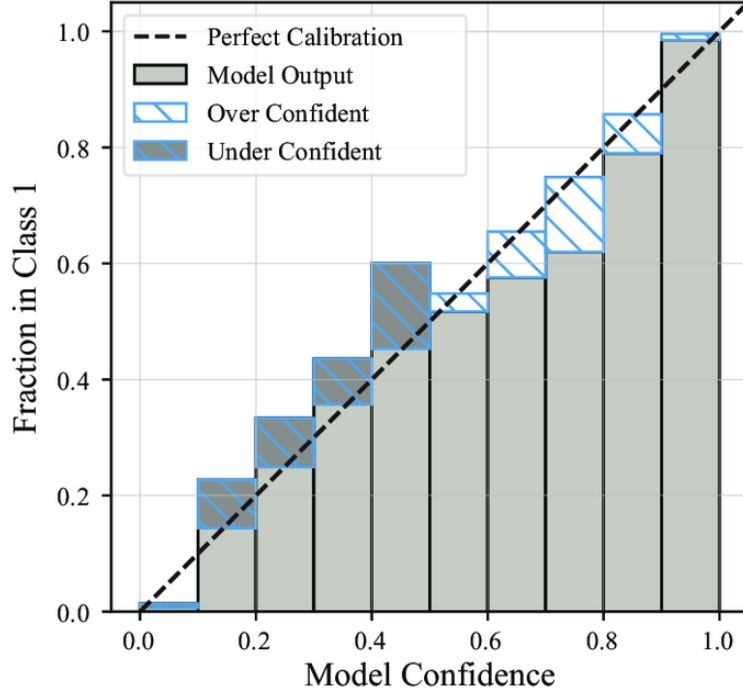


Figure 2.8: Reliability diagram illustrating the calibration of a model. The dashed line represents perfect calibration, where confidence matches accuracy. Bars above the diagonal indicate underconfidence, while those below indicate overconfidence.

Brier Score. The Brier Score is another widely used metric for assessing the quality of probabilistic predictions, particularly in binary classification tasks, and can be extended to voxel-wise segmentation outputs. It measures the mean squared difference between the predicted probability and the true outcome. For a single prediction, it is defined as:

$$\text{Brier Score} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2 \quad (2.4)$$

where $\hat{p}_i \in [0, 1]$ is the predicted probability for the i -th sample (e.g., the softmax output for a voxel), and $y_i \in \{0, 1\}$ is the ground truth label.

A lower Brier Score indicates better calibrated and more accurate predictions, as it reflects not only whether the prediction is correct, but also how close the predicted confidence is to the actual outcome. Unlike the ECE, which focuses on calibration discrepancy across confidence bins, the Brier Score provides a more holistic measure that combines both calibration and accuracy into a single value.

For instance, a perfectly calibrated model that always predicts 0.5 in a balanced dataset will have low ECE but a relatively high Brier Score, due to lack of resolution. Conversely, a well-calibrated and high-resolution model will score low on both metrics.

In the context of segmentation, the Brier Score complements the ECE by adding sensitivity to both overconfident incorrect predictions and underconfident correct ones. While ECE aggregates calibration discrepancies bin-wise, the Brier Score penalizes each miscalibrated voxel proportionally to the square of its error, making it particularly useful for fine-grained performance analysis.

Negative Log Likelihood (NLL). While originally designed as a loss function for training probabilistic models, the NLL also serves as a calibration-sensitive metric. A high NLL indicates that the predicted probability distributions poorly reflect the true outcome distribution. Unlike ECE, it penalizes overconfident wrong predictions more heavily.

2.7. Methods for Uncertainty Quantification in Deep Learning for Medical Image Segmentation

To quantify the uncertainties in the context of image segmentation, a variety of techniques have been developed, each with different trade-offs in terms of expressiveness, computational cost, and ease of integration. These methods can be broadly grouped into the following categories:

- **Bayesian Neural Networks (BNNs):** These models represent network weights as probability distributions rather than deterministic values. Posterior inference is approximated using techniques like variational inference or sampling. Despite being theoretically sound, BNNs are often impractical in large-scale segmentation due to high computational cost [4].
- **Monte Carlo Dropout (MC Dropout):** A widely used approximation to Bayesian inference, MC Dropout performs multiple stochastic forward passes with dropout layers enabled at test time. The variance of these predictions is used as a proxy for epistemic uncertainty. It is simple to implement and computationally efficient, making it a popular choice in medical imaging [7].
- **Deep Ensembles:** This approach trains multiple models independently with different initializations. Prediction variance across the ensemble reflects epistemic uncertainty, while the per-model variance captures aleatoric uncertainty. Ensembles

tend to be robust and well-calibrated but require significantly more resources for training and inference [16].

- **Test-Time Augmentation (TTA):** TTA applies various augmentations (e.g., flipping, rotation, noise) to the input at inference, producing multiple predictions. The diversity among these reflects aleatoric uncertainty. TTA is model-agnostic and can be combined with other approaches like MC Dropout or ensembles [26].
- **Evidential Deep Learning:** This framework encodes uncertainty by interpreting model outputs as parameters of a Dirichlet distribution over class probabilities. It enables a single forward pass to jointly estimate prediction and uncertainty, making it efficient and interpretable [27].
- **Generative Models and Structured Uncertainty:** Advanced models such as probabilistic U-Nets or stochastic segmentation networks can generate diverse, spatially coherent predictions, modeling both local and global uncertainties. These methods enable structural uncertainty estimates at the lesion- or case-level and are particularly relevant in clinical applications where object-level confidence is more informative than voxel-wise uncertainty [3].

Each of these methods plays a distinct role in capturing different sources of uncertainty. In practice, hybrid approaches—e.g., combining MC Dropout with TTA or deep ensembles—are often used to exploit complementary strengths. Regardless of the method, the goal remains the same: to augment the predictive power of deep learning models with meaningful estimates of confidence, thereby aligning AI behavior with clinical expectations for safety, transparency, and reliability.

2.7.1. Deep Ensemble for Uncertainty Estimation

One of the most effective and conceptually simple approaches for uncertainty estimation in deep learning is the use of *Deep Ensembles*. This method relies on training multiple instances of a neural network model—each initialized with different random seeds and possibly exposed to different data shufflings—so that their predictions form a diverse collection of hypotheses about the output. At inference time, the final prediction is typically computed by aggregating the outputs (e.g., via averaging), and the dispersion among these outputs is used as a proxy for epistemic uncertainty [16].

Definition and Rationale. An ensemble in machine learning is a collection of models whose outputs are combined to make a final prediction. In the context of deep learning, a **Deep Ensemble** refers to a set of neural networks $f_{\theta_1}, \dots, f_{\theta_M}$ trained independently

on the same task but initialized with different parameters. This strategy leverages the observation that neural networks, due to their non-convex loss landscapes, can converge to different local minima. By capturing these variations, ensembles can implicitly represent multiple modes of the posterior distribution over functions, making them powerful tools for capturing model (epistemic) uncertainty [34].

Formally, given an input x , the predictive distribution of the ensemble is given by:

$$p(y|x, \mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^M p(y|x, \theta_m), \quad (2.5)$$

where $\mathcal{D}$ is the training dataset, and θ_m are the parameters of the m -th model in the ensemble. The variability among the predictions across models—especially in cases where the data is sparse or ambiguous—can be interpreted as a reflection of epistemic uncertainty, as it reflects how sensitive the model’s outputs are to its parameters.

Uncertainty Estimation via Ensemble Statistics. In segmentation tasks, each model in the ensemble generates a voxel-wise probability distribution. Aggregation is typically performed by averaging the softmax outputs of the M models for each voxel. Let $p_i^{(m)}$ be the predicted class probabilities at voxel i from model m ; then the ensemble mean is:

$$\bar{p}_i = \frac{1}{M} \sum_{m=1}^M p_i^{(m)}. \quad (2.6)$$

A widely adopted approach to estimating epistemic uncertainty in this context is to compute the **variance across ensemble members**:

$$\text{Var}_i = \frac{1}{M} \sum_{m=1}^M \|p_i^{(m)} - \bar{p}_i\|^2. \quad (2.7)$$

This variance provides a quantitative measure of disagreement among the individual models, highlighting regions where the ensemble is less confident due to insufficient or ambiguous training data.

Benefits and Practical Considerations. The Deep Ensemble approach offers several compelling advantages. Firstly, it is conceptually straightforward and does not require any modification to the underlying network architecture or training objective. Secondly,

ensembles provide robust and well-calibrated uncertainty estimates [16, 21]. In addition, they often enhance the base model’s predictive performance by mitigating overfitting and averaging out individual model biases.

However, these benefits come at a computational cost. Training and storing M independent models increases memory usage and runtime proportionally with M , which may not be feasible in all deployment settings. Despite this, recent studies have shown that even relatively small ensembles (e.g., $M = 5$) suffice to obtain most of the uncertainty and accuracy benefits [6].

Use in Medical Image Analysis. Deep Ensembles have been successfully adopted in various medical imaging applications, particularly where model interpretability and reliability are essential. Tasks such as brain tumor segmentation, organ delineation, and lesion detection have benefited from ensemble-based uncertainty estimation [19, 25]. In these settings, ensemble models help to detect and flag unreliable predictions, thus supporting clinical quality assurance and decision-making.

In summary, Deep Ensembles offer a highly practical and empirically effective framework for uncertainty estimation in deep learning. Their ability to capture epistemic uncertainty without requiring complex modifications to model design or training processes makes them especially valuable in high-stakes domains like medical image analysis. While computationally demanding, their benefits in terms of reliability, robustness, and trustworthiness often justify the additional cost in research and clinical deployments.

2.7.2. Monte Carlo Dropout for Uncertainty Estimation

Monte Carlo Dropout (MC Dropout) is a widely adopted technique for estimating uncertainty in deep learning models, particularly valued for its simplicity and compatibility with standard training pipelines. Originally introduced by Gal and Ghahramani [7], the method reinterprets dropout—a commonly used regularization technique—as a form of approximate Bayesian inference, enabling the estimation of model uncertainty without altering the underlying architecture or training objective.

Definition and Rationale. In conventional deep learning, dropout is applied during training to randomly deactivate a subset of neurons in a layer, thereby reducing overfitting and promoting robustness. Typically, dropout is disabled at inference time, and the full network is used to produce deterministic predictions. However, Gal and Ghahramani proposed maintaining dropout active during inference and performing multiple forward passes with different random dropout masks. This results in a distribution of outputs

for each input sample, which can be interpreted as samples from the posterior predictive distribution of a Bayesian model [8].

Formally, let x^* be a test input and $f_{\hat{\omega}_t}(x^*)$ denote the output of the model under dropout mask t at inference. Performing T stochastic forward passes yields a set of predictions $\{f_{\hat{\omega}_1}(x^*), \dots, f_{\hat{\omega}_T}(x^*)\}$, which can be used to compute the expected output and quantify predictive uncertainty:

$$\mathbb{E}[y^*] \approx \frac{1}{T} \sum_{t=1}^T f_{\hat{\omega}_t}(x^*), \quad (2.8)$$

$$\text{Var}[y^*] \approx \frac{1}{T} \sum_{t=1}^T f_{\hat{\omega}_t}(x^*)^2 - (\mathbb{E}[y^*])^2. \quad (2.9)$$

Here, the variance serves as a proxy for **epistemic uncertainty**, capturing the variability induced by the model’s uncertain parameters.

Uncertainty Estimation via Multiple Forward Passes. In the context of image segmentation, the network produces voxel-wise probability distributions across the classes. With MC Dropout, each of the T stochastic forward passes generates a distinct softmax prediction map. The voxel-wise average of these predictions provides the final segmentation estimate, while the voxel-wise variance quantifies the epistemic uncertainty.

Let $p_i^{(t)}$ be the predicted class probabilities at voxel i from the t -th forward pass. The mean prediction is:

$$\bar{p}_i = \frac{1}{T} \sum_{t=1}^T p_i^{(t)}, \quad (2.10)$$

and the associated uncertainty at voxel i is estimated as:

$$\text{Var}_i = \frac{1}{T} \sum_{t=1}^T \|p_i^{(t)} - \bar{p}_i\|^2. \quad (2.11)$$

This variance indicates how consistent the model predictions are across different dropout-induced subnetworks. High variance typically signals areas where the model is uncertain due to insufficient exposure to similar patterns during training.

Benefits and Practical Considerations. MC Dropout presents numerous practical advantages. It is architecture-agnostic and requires no modifications to the training pipeline—only the inference procedure needs to be adapted to perform multiple stochastic passes. This makes MC Dropout especially attractive in scenarios where retraining a model is costly or infeasible. Additionally, it introduces only a moderate increase in computational load during inference, which can be controlled via the number of samples T [15].

Despite these strengths, the method also has limitations. The quality of the uncertainty estimates depends on the dropout rate and the number of inference samples. Moreover, as a variational approximation, MC Dropout may not fully capture the true posterior over the weights, leading to potential underestimation of uncertainty, particularly in the presence of strong out-of-distribution inputs [21]. Also, unlike ensemble-based methods, all forward passes in MC Dropout originate from a single trained model, potentially limiting the diversity of the sampled hypotheses.

In conclusion, MC Dropout represents a practical and theoretically grounded method for estimating uncertainty in deep learning. Its ease of integration, low computational overhead, and strong empirical performance have established it as a baseline approach in many uncertainty-aware medical image analysis pipelines. While not without limitations, it remains an essential tool in the development of robust and interpretable AI systems for healthcare.

2.7.3. Test-Time Augmentation for Uncertainty Estimation

Test-Time Augmentation (TTA) is a simple yet effective method for estimating uncertainty in deep learning models. Unlike traditional data augmentation techniques used during training to improve generalization, TTA is applied at inference time. The central idea is to introduce controlled variability into the test input—through a set of reversible transformations—and then analyze the variability in the model’s predictions across these transformed inputs. This approach allows the estimation of uncertainty without requiring architectural changes or retraining, making it highly attractive in clinical and real-time settings [20].

Definition and Rationale. The TTA process consists of four main steps. First, a set of deterministic augmentations (e.g., flipping, rotation, scaling, noise injection) is applied to the test image, generating N augmented versions of the input. The model is then used to predict a segmentation map for each augmented input. These predictions are then *de-*

augmented—i.e., transformed back into the original input space—to align all predictions spatially. Finally, the de-augmented predictions are aggregated, and their variability is analyzed to quantify the model’s uncertainty [30].

Formally, let x be an input image and $\mathcal{T} = \{t_1, \dots, t_N\}$ a set of deterministic augmentations. Each augmented version is $x_n = t_n(x)$, and the corresponding prediction is $y_n = f_\theta(x_n)$. The prediction is then de-augmented as $y_n^{\text{inv}} = t_n^{-1}(y_n)$ to bring it back to the original image space. The set $\{y_1^{\text{inv}}, \dots, y_N^{\text{inv}}\}$ is then used for aggregation and uncertainty estimation.

Uncertainty Estimation via TTA Variability. Once all de-augmented predictions are aligned, the model’s output can be taken as the mean prediction:

$$\bar{p}_i = \frac{1}{N} \sum_{n=1}^N p_i^{(n)}, \quad (2.12)$$

where $p_i^{(n)}$ is the predicted class probabilities at voxel i from the n -th augmented and de-augmented input. The associated voxel-wise variance is computed as:

$$\text{Var}_i = \frac{1}{N} \sum_{n=1}^N \|p_i^{(n)} - \bar{p}_i\|^2. \quad (2.13)$$

This variance captures how sensitive the model’s output is to minor changes in the input—effectively measuring **aleatoric uncertainty**, i.e., uncertainty arising from inherent noise or ambiguity in the data [30].

TTA can also be combined with MC Dropout to capture both epistemic and aleatoric uncertainty. In such cases, multiple stochastic forward passes are performed for each augmented input, and the total variability is decomposed into its two components.

Benefits and Practical Considerations. TTA offers several advantages. It is model-agnostic and does not require retraining, making it compatible with any pre-trained model. It also has a low implementation barrier, as most modern deep learning libraries provide efficient transformation tools. From a probabilistic perspective, TTA provides a non-parametric approximation of input-conditioned uncertainty by assessing the stability of model outputs under small input perturbations [20].

However, TTA introduces additional computational cost during inference, as N predictions must be made for each input. The trade-off between uncertainty estimation quality

and efficiency depends on the number and type of augmentations used. Moreover, care must be taken to ensure that augmentations are reversible and clinically valid—e.g., horizontal flipping may not be acceptable for anatomically asymmetric organs like the heart or liver.

In conclusion, Test-Time Augmentation is a powerful and flexible method for quantifying aleatoric uncertainty in medical image segmentation. Its simplicity, adaptability, and compatibility with existing models make it a valuable tool in clinical workflows where robustness and reliability are paramount.

2.7.4. Checkpoints Ensemble for Uncertainty Estimation

The *Checkpoints Ensemble* method is a recent and highly efficient technique for estimating uncertainty in deep learning models, particularly tailored to self-configuring architectures such as **nnU-Net**. Unlike traditional Bayesian approaches that require architectural changes or significant computational resources, this method leverages the natural behavior of stochastic gradient descent (SGD) to extract posterior samples of network weights during training [36].

Conceptual Foundations. During the training of neural networks using SGD, especially under non-converging learning schedules, model weights naturally fluctuate around local optima. This fluctuation can be interpreted as a stochastic traversal of the posterior distribution in weight space [18]. Building on this idea, the Checkpoints Ensemble technique proposes to periodically save network checkpoints from later training epochs—particularly after the model has stabilized—to approximate samples from the posterior $p(w|\mathcal{D})$. By aggregating predictions from these diverse checkpoints, it becomes possible to capture epistemic uncertainty without modifying the architecture or introducing sampling noise at inference [36].

Method Description. Let $\mathcal{D}$ denote the training dataset, and w_t be the network weights at epoch t . After an initial burn-in phase (e.g., the first γT epochs), the model begins posterior sampling by collecting weights $\{w_t\}_{t=\gamma T}^T$. These checkpoints are then used for Monte Carlo approximation of the predictive distribution:

$$p(y^*|x^*, \mathcal{D}) \approx \frac{1}{n} \sum_{i=1}^n p(y^*|x^*, w_{t_i}), \quad (2.14)$$

where n is the number of selected checkpoints, and $p(y^*|x^*, w_{t_i})$ is the model prediction

at weights w_{t_i} . Since each checkpoint corresponds to a slightly different configuration due to SGD noise, their aggregation can reflect uncertainty in areas of low model confidence.

Multi-Modal Posterior Sampling. To enhance posterior diversity, the authors propose a *cyclical learning rate schedule*, where the training is divided into M cycles. At the beginning of each cycle, the learning rate is reset to a high value, encouraging the model to escape from previous minima and explore new regions of the weight space. Within each cycle, checkpoints are collected after a warm-up phase and used as independent posterior modes. The resulting ensemble therefore captures both local and global uncertainty by combining intra-mode and inter-mode variability [36].

In summary, Checkpoints Ensemble combines the strengths of Bayesian uncertainty estimation with the practical ease of standard training pipelines. Its applicability to nnU-Net and improved performance on out-of-domain data make it a valuable tool for trustworthy segmentation in medical imaging.

3 | Uncertainty and Calibration in Medical Image Segmentation

This chapter provides a comprehensive overview of the main research directions in this domain, with particular emphasis on uncertainty estimation and calibration techniques that are both theoretically grounded and clinically relevant.

While Chapter 2 introduced the basic concepts of uncertainty and calibration, from a more technical point of view. Here we delve into how these ideas have been operationalized in recent literature, especially in the context of semantic segmentation.

The chapter begins by focusing on calibration techniques, highlighting their role in aligning confidence scores with prediction reliability. We then review the main families of uncertainty estimation approaches—ranging from sampling-based methods like deep ensembles and test-time augmentation, to learned and evidential frameworks—and analyze their strengths and limitations when applied to medical image segmentation. Finally, we consider the problem of uncertainty visualization and its integration into clinical workflows, which is essential for enabling human-machine collaboration and for fostering the adoption of AI systems in healthcare.

Together, these sections outline the state of the art in trustworthy segmentation pipelines, forming the scientific basis for the experimental design and methods proposed in the following chapters.

3.1. Calibration of Predictive Confidence in Medical Image Segmentation

As explained in section 2.6, one of the fundamental limitations of deep learning models in critical applications such as medical imaging is their tendency to produce overconfident predictions, even in situations of poor data quality or under distributional shift. A well-calibrated model is one whose confidence scores accurately reflect the model's expected

accuracy: for example, among all predictions to which the model assigns a confidence of 0.8, approximately 80% should be correct.

As demonstration the seminal work of Guo et al. [10] highlighted that modern deep neural networks, despite achieving high accuracy, often suffer from miscalibration. In their study, they proposed **temperature scaling** as a simple yet effective post-hoc calibration method, which consists in dividing the logits by a scalar temperature parameter before applying the softmax function. The optimal temperature is typically found on a validation set by minimizing a calibration error metric such as the Expected Calibration Error (ECE). This approach preserves the classification accuracy but smooths the predicted probabilities, making them more representative of actual likelihoods. Temperature scaling has since become a standard baseline in calibration research.

In the context of medical image segmentation, calibration has gained traction as a necessary step to increase the trustworthiness of deep learning systems. Mehrtash et al. [19] explored the calibration properties of segmentation models by evaluating the discrepancy between predicted confidence maps and actual performance. Their work systematically analyzed how models trained for prostate segmentation and brain tumor segmentation tend to be miscalibrated, especially near object boundaries or in regions of ambiguous tissue. By applying temperature scaling and ensemble methods post-hoc to the segmentation logits, they demonstrated a clear reduction in miscalibration as measured by the ECE, without deteriorating the segmentation quality in terms of Dice score.

A more recent and comprehensive contribution is provided by Rousseau et al. [24], who reviewed and benchmarked a range of post-hoc calibration methods for segmentation, including **temperature scaling**, **histogram binning**, and **isotonic regression**. Unlike previous studies, they evaluated calibration not only at the voxel level but also for **whole-region** predictions, accounting for clinical scenarios where segmentation volumes rather than pixel-wise labels are used for diagnosis or treatment planning. The study revealed that temperature scaling remains one of the most effective and robust approaches, particularly when applied separately per class in multi-class segmentation tasks. The authors emphasized the necessity of selecting proper calibration metrics depending on the clinical objective—for example, per-structure reliability estimation in radiotherapy planning versus global uncertainty estimates for triaging.

Beyond post-hoc techniques, several works have attempted to integrate calibration-awareness directly **into the training process**. Dawood et al. [5] proposed an uncertainty-aware training strategy for cardiac MR image classification, which includes a regularization term that penalizes miscalibrated predictions. Their method showed improved calibration and

reliability, particularly in underrepresented pathological cases, and helped in reducing false positives in binary classification without sacrificing sensitivity. Although not focused on segmentation, their approach suggests a promising avenue for future work in directly incorporating calibration constraints into segmentation losses.

Lambert et al. [17] further framed calibration within the broader scope of trustworthy AI. They provided a structured review of calibration techniques used in deep learning for medical image analysis and highlighted the connections between calibration, uncertainty estimation, and clinical applicability. In their survey, they discussed methods based on Dirichlet distributions and evidential deep learning that aim to model prediction confidence more faithfully. Moreover, they observed that well-calibrated models are more effective when used in conjunction with uncertainty estimation pipelines, particularly when decisions are deferred to human experts in high-uncertainty regions. The integration of calibration metrics into system evaluation was presented not merely as a technical addition, but as a requirement for ethical and interpretable clinical deployment.

Zhao et al. [36] brought calibration methods into the context of the nnU-Net framework. Their work focused on efficient Bayesian uncertainty estimation and its interplay with calibration. By evaluating the ECE and reliability diagrams of their modified nnU-Net models, they provided evidence that certain stochastic inference mechanisms, such as Monte Carlo dropout and deep ensembles, improve calibration implicitly but still benefit from explicit post-hoc adjustment via temperature scaling. This aligns with the broader consensus in the literature that Bayesian methods, while providing richer uncertainty estimates, do not guarantee well-calibrated confidence outputs.

Taken together, these studies confirm that calibration plays a crucial role in enhancing the reliability of deep learning models for medical imaging tasks. Whether through post-hoc techniques like temperature scaling or integrated uncertainty-aware training, calibration allows confidence estimates to become a reliable indicator of prediction correctness. This is essential not only for model interpretability and user trust, but also for the safe adoption of AI tools in high-stakes environments such as diagnosis, treatment planning, and surgery assistance.

3.2. Uncertainty Estimation Methods in Medical Imaging

Several approaches have been developed in recent years to incorporate uncertainty estimation into deep learning models for medical image analysis, and some of them were widely

technically described and discussed in section 2.7. These methods differ in theoretical foundation, computational cost, and practical implications, but they all aim to provide insight into the confidence of a model’s predictions. Within the literature, a number of techniques have emerged as particularly influential and widely adopted.

A prominent class of approaches is based on **Monte Carlo dropout**, where dropout layers are kept active at inference time to generate multiple stochastic forward passes. This method, introduced as a practical approximation to Bayesian inference in deep neural networks, has been effectively applied in various clinical tasks. In the domain of medical image segmentation, for example, it has been used to estimate voxel-wise uncertainty maps that highlight regions where predictions are less reliable [30]. Similarly, [25] employ this technique to evaluate structure-wise reliability in brain segmentation, demonstrating how uncertainty maps can be leveraged for automatic quality control in large-scale neuroimaging datasets.

Another approach that has gained considerable attention is the use of **deep ensembles**. Here, multiple models are trained independently with different random initializations or data shuffling, and the variability in their outputs is used to estimate epistemic uncertainty. The work of [16] outlines the theoretical motivation behind this method and demonstrates its robustness across a variety of tasks. In the medical domain, ensembles have been successfully applied to both classification and segmentation, as discussed in the broader reviews of [1] and [37], which highlight their effectiveness in handling out-of-distribution samples and in improving the reliability of automated diagnostic tools.

Closely related to this idea is the concept of **checkpoint ensembles**, in which different snapshots of the same model taken at various points during training are used instead of independently trained networks. While this technique is computationally more efficient than training separate models from scratch, it can still capture a meaningful estimate of uncertainty, especially when combined with other mechanisms such as stochastic data augmentation or regularization. Applications of checkpoint ensembles are referenced in broader surveys like [37], where they are positioned as a practical compromise between single-model simplicity and full ensemble robustness.

An additional strategy involves the use of **test-time data augmentation**, where the input image is perturbed multiple times at inference and the model’s responses to these variations are aggregated. This technique exploits the model’s sensitivity to input-level changes to derive a proxy for predictive uncertainty. As demonstrated in [30], this approach is particularly useful for identifying aleatoric uncertainty and has shown effectiveness in both classification and segmentation tasks. In some implementations, test-time

augmentation is also used in combination with MC dropout or ensemble predictions to enhance the resolution and reliability of uncertainty maps.

In contrast to sampling-based methods, there has been a growing interest in deterministic techniques, particularly those derived from **evidential deep learning**. These methods avoid multiple forward passes by modeling prediction uncertainty through evidence-based distributions, such as the Dirichlet distribution over class probabilities. [12] provide an extensive overview of evidential approaches, noting their advantages in terms of computational efficiency and robustness to out-of-distribution data. By interpreting network outputs as subjective opinions rather than fixed estimates, evidential methods offer a principled framework for handling both aleatoric and epistemic uncertainty without the need for Bayesian sampling.

Across the surveyed literature, the choice of method often reflects a trade-off between computational cost and expressive power. While sampling-based techniques like Monte Carlo dropout and deep ensembles provide detailed uncertainty estimates, they are typically more expensive to deploy. Deterministic approaches, on the other hand, offer practical benefits in real-time scenarios. As noted in comprehensive reviews such as [37] and [17], integrating these techniques into clinical workflows remains a key challenge, and the field continues to evolve with the aim of producing uncertainty estimates that are not only theoretically grounded but also actionable in practice.

3.2.1. Visualization of Uncertainty in Medical Image Segmentation

The effective visualization of uncertainty in medical image segmentation is essential for clinical integration of deep learning systems. While predictive metrics and scalar evaluations are crucial for benchmarking algorithms, they do not always convey interpretable information to human users—particularly clinicians—who require intuitive and actionable representations of model confidence. Therefore, visual tools, most notably *uncertainty heatmaps*, have been adopted to bridge this gap, offering spatial cues about the reliability of each predicted voxel [37].

From Confidence Maps to Entropy Maps. Most segmentation models produce a voxel-wise softmax probability distribution over K possible classes. The *confidence map* is typically constructed by extracting the highest softmax value at each voxel:

$$\text{Confidence}(x_i) = \max_k p_{i,k}, \quad (3.1)$$

where $p_{i,k}$ is the predicted probability for class k at voxel i . While this gives a rough indication of prediction certainty, it only reflects the model’s commitment to its top choice, not the overall uncertainty across all classes.

To better represent the predictive distribution, the *entropy* of the softmax output is often computed:

$$\text{Entropy}(x_i) = - \sum_{k=1}^K p_{i,k} \log p_{i,k}. \quad (3.2)$$

This scalar captures the degree of disorder or ambiguity in the prediction. High entropy suggests that the model is unsure and is assigning similar probabilities to multiple classes, whereas low entropy indicates confident predictions with clear class dominance.

Binary vs Multiclass Cases. In the binary segmentation setting ($K = 2$), the entropy simplifies to:

$$\text{Entropy}(x_i) = -p_i \log p_i - (1 - p_i) \log(1 - p_i), \quad (3.3)$$

where p_i is the predicted probability of the positive class. This function reaches its maximum at $p_i = 0.5$ and its minimum at $p_i = 0$ or 1 , thus providing an intuitive scale for measuring uncertainty.

In multiclass segmentation ($K > 2$), entropy increases with the number of classes and is often normalized by $\log K$:

$$\text{NormalizedEntropy}(x_i) = \frac{- \sum_{k=1}^K p_{i,k} \log p_{i,k}}{\log K}, \quad (3.4)$$

ensuring the output values are bounded between 0 and 1. This normalization facilitates consistent visualization across datasets with varying class counts.

Visualization Techniques. Entropy or confidence maps are typically displayed as *heatmaps* overlaid on the original image or segmentation mask. These maps allow clinicians to visually inspect areas where the model exhibits ambiguity—e.g., near lesion boundaries or in regions affected by noise or motion artifacts. High-uncertainty regions are usually highlighted using warmer colors (e.g., red or yellow), guiding expert attention.

Some works propose composite visualizations that integrate predictions, uncertainty, and anatomical priors. For example, Moshkov et al. [20] demonstrated how test-time augmentation can generate variability maps that align with clinically ambiguous regions. Similarly, Jungo et al. [14] used voxel-wise uncertainty derived from MC Dropout to identify segmentation boundaries where inter-observer disagreement was high.

These visual tools have several clinical implications:

- They can serve as a quality control mechanism, allowing automatic flagging of cases requiring manual review.
- They support expert annotation by highlighting regions where multiple model hypotheses exist.
- They enable visual triaging in referral workflows, improving efficiency in large-scale screening scenarios.

Heatmaps derived from entropy have been used in numerous studies as a means of expressing predictive uncertainty, especially in deep Bayesian models and ensemble techniques [25, 36]. Notably, entropy-based visualization has proven more reliable and interpretable than raw softmax confidence in the presence of class imbalance and domain shift [21].

Despite their utility, entropy maps have limitations. They do not explicitly distinguish between aleatoric and epistemic uncertainty, unless the predictive distributions are decomposed via multiple models (as in ensembles) or augmented inputs (as in TTA). Moreover, the interpretability of these maps relies heavily on consistent calibration of the model outputs. Ongoing research is exploring how to refine these visualizations using hierarchical uncertainty decomposition, information-theoretic criteria, and clinical priors [30, 37].

In conclusion, uncertainty heatmaps—particularly those based on entropy—are indispensable for translating abstract predictive confidence into actionable clinical insights. Their ability to highlight failure-prone regions fosters safer, more transparent, and more trustworthy integration of AI into medical workflows.

4 | Dataset and Experiments

This chapter provides a detailed description of the experimental framework developed to investigate uncertainty estimation in medical image segmentation. It begins by introducing the dataset used in this study—comprising full-body CT scans of 45 patients—along with relevant characteristics such as voxel spacing, resolution heterogeneity, and the clinical significance of the annotated structures.

Building on this foundation, we describe the training strategy employed using the **nnU-Net v2** framework, including both standard cross-validation ensembles and a customized checkpoint-based training scheme designed to enable posterior sampling. The chapter then moves to the implementation of uncertainty estimation techniques, highlighting their integration into the nnU-Net pipeline and discussing the types of uncertainty they aim to capture.

Finally, we present the methods used for visualizing uncertainty maps, with particular emphasis on entropy-based approaches that allow intuitive interpretation of model confidence at the voxel level. The contents of this chapter serve as the methodological basis for the analyses and results discussed in the following sections.

4.1. Dataset

The private dataset used for the experimental part of this thesis consists of three-dimensional computed tomography scan (CT) scans from a total of 45 patients. Each volume in the dataset represents a complete CT scan of a single patient, annotated with expert-generated manual segmentations identifying the lymphatic system.

The aim of this segmentation is to delineate the anatomical regions occupied by the lymphatic structures, which are of particular clinical interest for several diagnostic and treatment planning tasks. The annotations were performed by domain experts and are considered the ground truth for training and evaluation purposes. These expert labels are essential in supervised deep learning pipelines, especially for segmentation tasks where pixel-level precision is crucial.

Volume characteristics All volumes are stored in the NIfTI file format, a standard in neuroimaging and medical image analysis. The spatial dimensions of the volumes are consistent in the axial and sagittal planes (512×512 pixels), while the number of slices in the coronal direction varies across patients. The diversity in slice count reflects the differences in acquisition protocols or anatomical coverage.

Figure 4.1 shows the distribution of the dataset volumes grouped by their shape. As can be observed, the slice count ranges from 192 to 398 voxels in the axial direction, with several values recurring more frequently—particularly 219, 232, and 243 slices. This variability introduces additional complexity during preprocessing and network input standardization, as all volumes must be resampled or padded to meet the fixed input requirements of the model architecture.

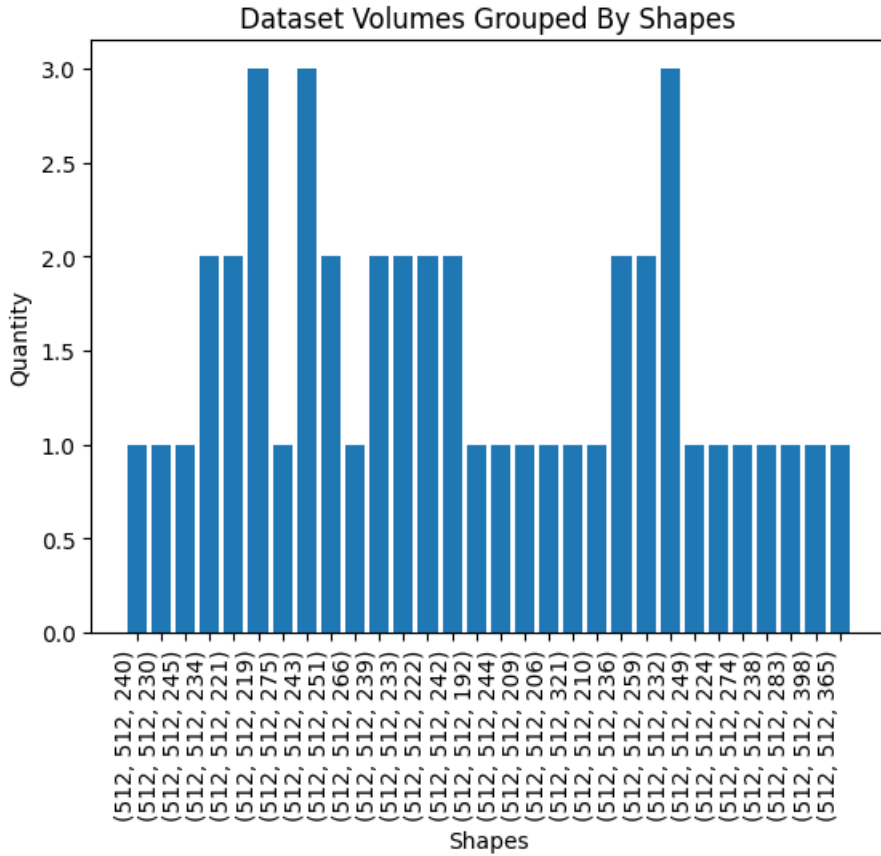


Figure 4.1: Histogram showing the distribution of dataset volumes grouped by shape (width, height, depth). All volumes have fixed axial dimensions of 512×512 pixels, while the number of slices in the coronal direction varies.

Voxel spacing The physical resolution of each voxel, defined by the spacing between voxels in each spatial direction, is not uniform across the dataset. The in-plane voxel

size (X and Y axes) varies between approximately 0.976 mm and 1.5625 mm, while the inter-slice spacing (Z-axis) assumes discrete values of 3.0 mm, 5.0 mm, or 7.5 mm.

As shown in Figure 4.2, the most common voxel spacing in the dataset is (1.171875, 1.171875, 5.0) mm, which appears in 29 of the 45 volumes. Other spacing configurations occur less frequently but still contribute to the heterogeneity of the dataset. This variability can influence model training, especially in 3D convolutional networks that are sensitive to anisotropy in the spatial dimensions. To address this, all volumes were resampled by nnU-Net to a common target spacing, chosen as the median spacing across the dataset.

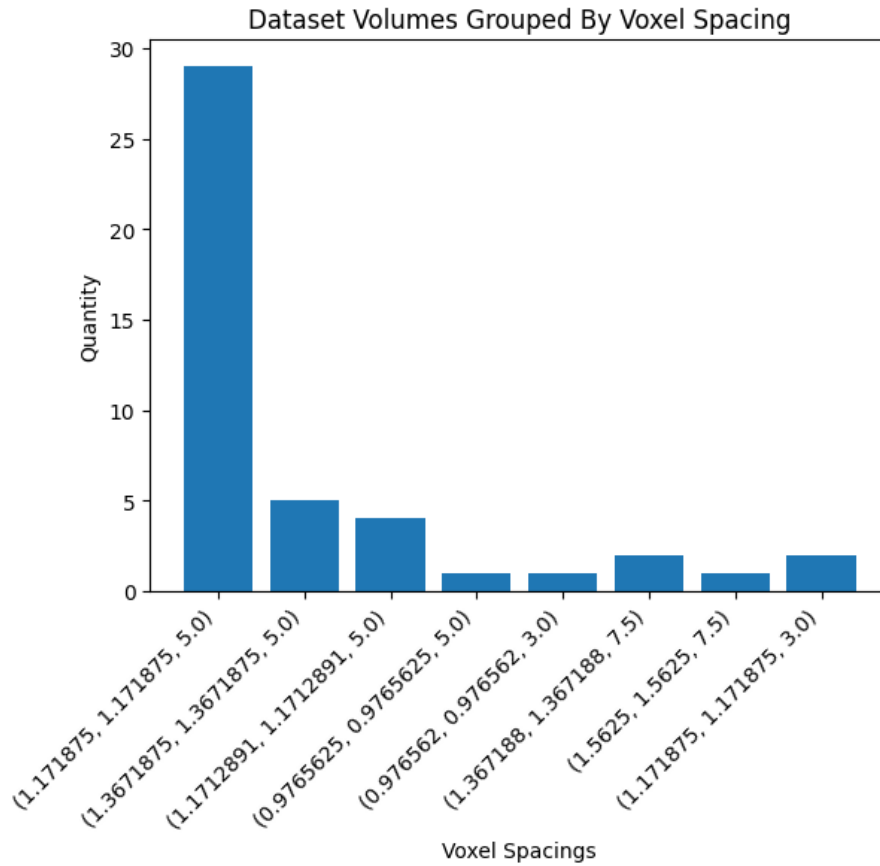


Figure 4.2: Histogram showing the number of dataset volumes grouped by voxel spacing configuration. The majority of scans share the same in-plane resolution of approximately 1.17 mm and a slice thickness of 5.0 mm.

Clinical relevance and challenges The lymphatic system is a complex and diffuse anatomical structure, often poorly contrasted in standard CT sequences. As such, accurate manual delineation is both time-consuming and prone to inter-observer variability. These challenges are reflected in the dataset, and they motivate the integration of uncertainty estimation in segmentation models. By explicitly modeling uncertainty, one

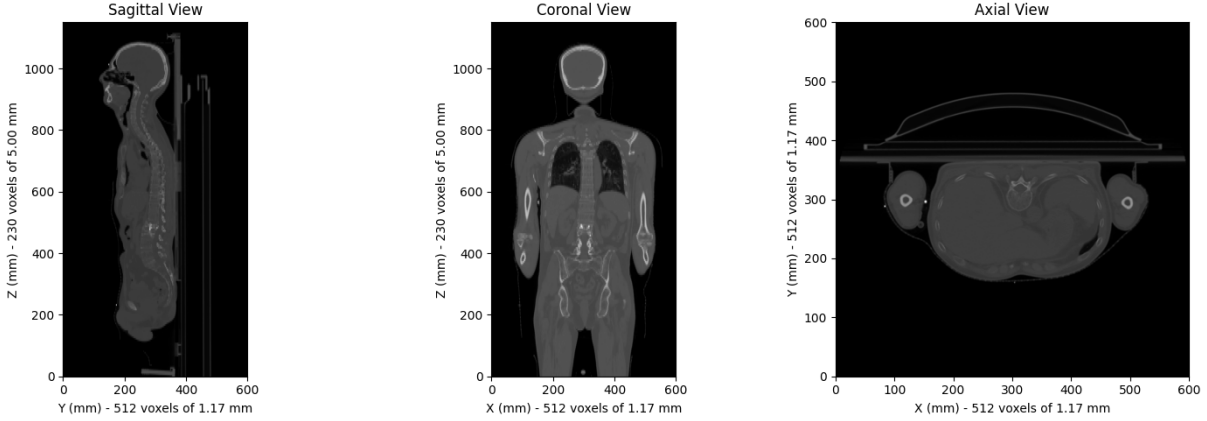


Figure 4.3: Example of a volume from the dataset displayed in the three anatomical planes: sagittal (left), coronal (center), and axial (right) views. The image corresponds to a randomly selected patient scan and illustrates the volumetric nature of the data along with the voxel spacing and resolution: 512×512 pixels in the axial plane, and a slice thickness of 5.00 mm in the Z-direction.

can better identify ambiguous regions and improve the trustworthiness of the automated predictions in clinical applications.

Although the dataset comprises a relatively small number of samples ($N = 45$), the risk of overfitting is mitigated by the use of the **nnU-Net** framework, which is specifically designed to adapt to medical imaging tasks with limited annotated data. **nnU-Net** dynamically configures preprocessing, architecture, training, and post-processing steps according to the characteristics of the dataset, making it a robust and effective solution even in low-data regimes.

4.2. Model and Training

All experiments conducted in this thesis are based on the **nnU-Net** framework [13]. The choice of nnU-Net was motivated by its ability to autonomously adapt its architecture, training scheme, and preprocessing pipeline to the properties of the dataset, making it particularly suitable for scenarios with limited annotated data—such as the one considered in this study.

Two distinct training configurations were explored:

- A standard **cross-validation ensemble**, using the default 3D full resolution trainer provided by the nnU-Net library.
- A custom **checkpoints ensemble** implementation, based on a cyclical learning rate

strategy and posterior sampling via SGD checkpoints, as proposed in [36].

The details of these configurations are reported in the following subsections.

4.2.1. Ensemble Training

The first model configuration leverages the default 5-fold cross-validation setup provided by nnU-Net. Out of the total 45 annotated volumes, 40 were used for training and validation. The remaining 5 were completely held out and reserved for final evaluation. The 5-fold training follows this procedure:

- The 40 training cases are divided into 5 equally sized folds.
- For each fold, the model is trained on 32 volumes and validated on the remaining 8.
- This produces 5 separate models trained on different data splits.

Each model was trained using the `3d_fullres` configuration, which was automatically selected by nnU-Net based on the dataset fingerprint. The framework also handles all required preprocessing steps (e.g., resampling, normalization, cropping), as well as architectural decisions (e.g., depth, number of filters) in a fully automatic fashion.

4.2.2. Checkpoints Ensemble Training

To incorporate model uncertainty estimation without modifying the network architecture or increasing inference time per sample, we implemented the *Checkpoints Ensemble* strategy described in [36]. This method utilizes weight checkpoints collected throughout training to approximate posterior samples, enabling uncertainty estimation through a lightweight Monte Carlo ensemble.

Custom Trainer. To implement this method, a new trainer class named `nnUNetTrainerMultiModalCycLR` was developed by extending the base trainer provided by nnU-Net. This trainer was saved as a Python file into the nnunet library to extend it for this purpose.

Training Behavior. The trainer divides the full training duration ($T = 1200$ epochs) into $M = 3$ cycles. Each cycle uses a cyclical learning rate schedule to promote exploration of different local minima. During the final 10 epochs of each cycle, model checkpoints are saved. These checkpoints are later used as individual models for ensemble inference, approximating samples from the posterior distribution $p(w|\mathcal{D})$.

Learning Rate Strategy. The learning rate at epoch t within a cycle is computed via a polynomial decay:

$$\alpha(t) = \begin{cases} \alpha_r & \text{if } t_c = 0, \\ \alpha_0 \left[1 - \frac{\min(t_c, \gamma T_c)}{T_c} \right]^\epsilon & \text{otherwise,} \end{cases}$$

where:

- $\alpha_r = 0.1$ is the initial restart learning rate,
- $\alpha_0 = 1\text{e-}2$ is the base learning rate,
- $\gamma = 0.8$, $\epsilon = 0.9$,
- $T_c = T/M$ is the number of epochs per cycle.

This schedule ensures exploration of distinct weight regions during each cycle, supporting a multimodal posterior approximation as proposed in [36].

Training Execution. In this case, just a unique fold were used on the 40-case training subset. The 5 test volumes were excluded from this process. In Figure 4.4 it is possible to

see the progresses during the training of this model showing the behaviour of the cyclical loss and learning rate updates described so far.

4.3. Implemented Methods for Uncertainty Estimation and Visualization

In this section, we provide a detailed overview of the uncertainty estimation and visualization methods that were implemented for medical image segmentation using deep learning. The theoretical foundations and the classification of uncertainty types—namely, epistemic and aleatoric uncertainty—have already been discussed in Section 2.6 and Section 2.7. Therefore, the focus here is on the practical aspects: how these methods were concretely implemented and integrated into the experimental pipeline based on the **nnU-Net v2** framework of this work.

Each of the following subsections presents a specific strategy for uncertainty estimation or visualization. For each method, we describe how it was operationalized, the type of uncertainty it addresses, and how it was evaluated in the context of 3D medical image segmentation.

It is important to note that Monte Carlo Dropout (MC Dropout) was deliberately excluded from the set of implemented techniques. Although this approach is often referenced in the literature as a practical approximation to Bayesian inference, it has been shown to significantly degrade segmentation performance when applied to high-resolution volumetric data. According to Fabian Isensee, one of the principal developer of **nnU-Net**, activating dropout layers during inference reduces the network’s effective capacity. For these reasons, MC Dropout was not considered suitable for our clinical-grade segmentation pipeline.

Given these limitations, the focus was placed on more stable and high-fidelity uncertainty estimation techniques. These include deep ensembles, test-time data augmentation, temperature scaling for confidence calibration, ensemble methods based on model checkpoints, and entropy-based visualizations. All methods were adapted to fit the **nnU-Net v2** framework and evaluated both quantitatively and qualitatively.

The following subsections provide an in-depth description of each method, including implementation details, uncertainty metrics, visualization strategies, and a critical discussion of their respective strengths and limitations in the context of uncertainty-aware medical image segmentation.

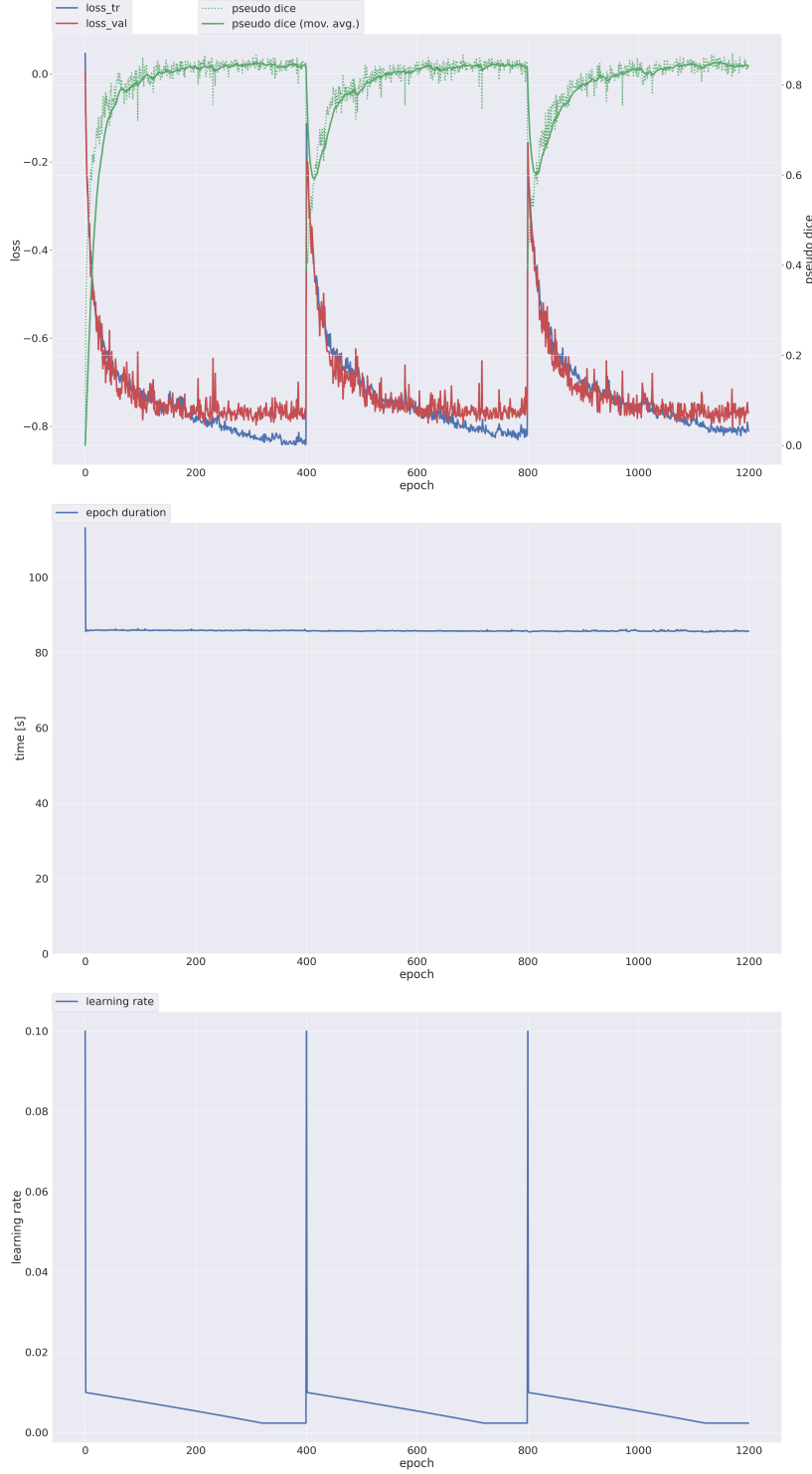


Figure 4.4: Training progress of the custom `nnU-NetTrainerMultiModalCycLR` used for the Checkpoints Ensemble method, shown for the unique fold 0. **Top:** Training and validation loss curves (left axis) and pseudo-Dice score on the validation set (right axis) over 1200 epochs. The plot highlights the effect of cyclical learning rates, with three distinct phases corresponding to the three learning cycles. After each cycle restart, a temporary degradation is followed by a rapid recovery and convergence, confirming that the optimizer escapes local minima and explores different modes in the posterior weight space. **Middle:** The duration of each training epoch in seconds, which remains stable throughout the training process, indicating consistent computational efficiency. **Bottom:** Learning rate schedule defined by the cyclical strategy. Every 400 epochs, the learning rate is reset to a high value ($\alpha_r = 0.1$) and then decays polynomially following Equation (4) in [36]. This behavior enables multi-modal posterior sampling by encouraging the optimizer to explore different regions of the weight landscape across training cycles.

4.3.1. Temperature Scaling for Calibration

The first step undertaken for extracting uncertainty from the trained models focused on evaluating and improving their calibration. To assess the initial calibration status, we employed *reliability diagrams* and computed the *Expected Calibration Error (ECE)* across all voxels of the test set. However, to avoid bias due to the predominance of background voxels—which typically exhibit high confidence and dominate volumetric datasets—we restricted the analysis to a more informative subregion. Specifically, we included only those voxels located within a maximum Euclidean distance of 24.5 mm from the ground truth segmentation of each patient.

To improve model calibration, we implemented **Temperature Scaling**, a post-hoc calibration technique that scales the logits by a constant temperature parameter T before applying the softmax function. The temperature acts as a smoothing factor: values $T > 1$ reduce model confidence (addressing overconfidence), while values $T < 1$ increase it (addressing underconfidence).

To facilitate practical use, modifications were made to the **nnU-Net v2** default inference pipeline. In particular, the inference script parser was extended to support a new argument `-T [value]`, allowing users to apply temperature scaling directly from the command line using the `nnunetv2_predict` command. Internally, this option ensures that all predicted logits are divided by the specified value T prior to the softmax activation, producing probability masks and subsequent binary segmentations accordingly.

Given its simplicity and minimal computational cost, we adopted an empirical evaluation approach. Using initially a single fold (fold 0) from the baseline ensemble trained with the default trainer (described in Section 4.2), we compared the reliability diagrams under different configurations: no temperature scaling, $T > 1$, and $T < 1$. The following comparison has been done also with all the other methods we are going to see later, and they all have achieved similar results.

Figure 4.5 presents a side-by-side comparison of the reliability diagrams obtained for these three settings. All diagrams aggregate voxel-wise predictions across all patients in the validation set, considering only those voxels within 24.5 mm from the ground truth as previously described.

From these diagrams, it is evident that the original model ($T = 1$) is significantly overconfident across most confidence bins, with observed accuracy falling well below predicted confidence. Applying temperature scaling with $T = 3.0$ (Figure 4.5c) noticeably improves calibration, with observed accuracy more closely tracking the ideal diagonal line.

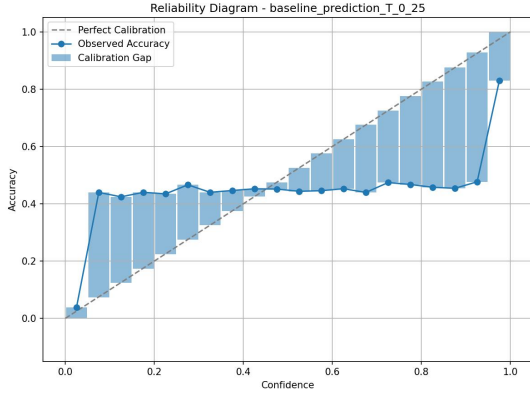
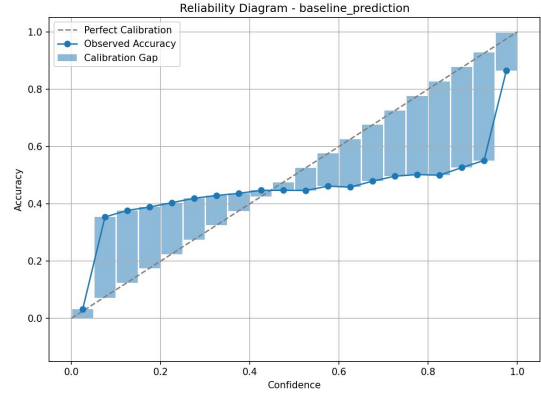
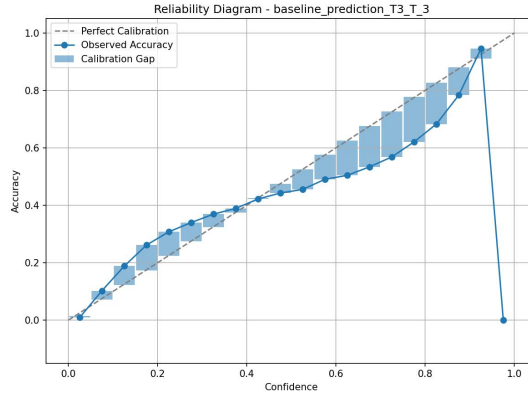
(a) $T = 0.25$ (b) $T = 1$ (No scaling)(c) $T = 3.0$

Figure 4.5: Reliability diagrams comparing the calibration of predictions from fold 0 of the baseline ensemble. Only voxels within 24.5 mm of the ground truth segmentations were considered. Each diagram aggregates predictions across all patients in the test set.

In contrast, applying a temperature of $T = 0.25$ (Figure 4.5a) leads to a more severe misalignment, amplifying overconfidence especially in the medium- to high-confidence bins.

Table 4.1: Comparison of Expected Calibration Error (ECE) across different temperature scaling settings. Values are computed over all voxels of the whole test set dataset within 24.5 mm of the ground truth segmentations.

Experiment	Temperature	ECE (%)
Baseline (no scaling)	$T = 1$	5.45
Underscaled	$T = 0.25$	6.23
Overscaled	$T = 3.0$	2.39

These qualitative observations are confirmed by the quantitative comparison of ECE values reported in Table 4.1. Once again, we observe that increasing the temperature leads to a substantial reduction in calibration error, whereas a temperature lower than 1 results in even poorer calibration.

This empirical analysis, supported by both qualitative (reliability diagrams) and quantitative (ECE) metrics, demonstrates the practical benefit of temperature scaling for model calibration. In particular, scaling the logits with a temperature greater than one effectively mitigates overconfidence and brings the model’s confidence estimates closer to the true observed accuracy, enhancing the reliability of uncertainty quantification in downstream tasks.

It is worth noting that while the temperature scaling analysis confirmed the benefits of applying $T > 1$, we deliberately chose not to explore values significantly higher than 3.0. This decision was motivated by the intent to avoid overly aggressive modifications to the model’s original confidence distributions. Excessive scaling may excessively flatten the softmax outputs, diminishing the model’s ability to express meaningful confidence differences between predictions. In the context of medical image segmentation—where precise probabilistic outputs are often crucial for downstream interpretation and decision-making—it was deemed more appropriate to pursue a conservative calibration strategy.

4.3.2. Deep Ensemble Implementation

Among the various strategies explored for uncertainty estimation in segmentation tasks, the deep ensemble approach was also implemented due to its empirical robustness and conceptual simplicity. In this specific application, the integration of deep ensembles was particularly straightforward thanks to the native support provided by the **nnU-Net v2** framework. Unlike other methods that require additional scripting or architectural modifications, ensemble inference is natively supported as a built-in feature within the library, and does not necessitate any custom code development.

The underlying idea of the ensemble method is to leverage multiple models, each trained with a different initialization seed (corresponding to the five folds of cross-validation), and aggregate their predictions to produce a more robust and uncertainty-aware output. In the **nnU-Net v2** pipeline, each fold corresponds to a model trained on a different subset of the training data, while being validated on a unique validation split. This diversity across models is exploited during inference to estimate epistemic uncertainty via prediction variance.

To run the ensemble inference, the command-line interface of **nnU-Net v2** was used. The

following command was issued to perform inference using all five folds:

```
command = [
    "nnUNetv2_predict",
    "-d", "Dataset004_with_test_set",
    "-i", input_dir,
    "-o", output_dir,
    "-f", "0", "1", "2", "3", "4",
    "-tr", "nnUNetTrainer",
    "-c", "3d_fullres",
    "-p", "nnUNetResEncUNetMPlans",
    "-T", str(T),
    "--save_probabilities"
]
```

This command instructs the framework to:

- use all five cross-validation folds (`-f 0 1 2 3 4`) for ensemble inference,
- apply temperature scaling with a user-specified temperature value (`-T`),
- save the resulting softmax probability maps for further uncertainty analysis (`-save_probabilities`).

The outputs from each fold, consisting of voxel-wise softmax probability maps, are internally averaged by the framework to yield a single final prediction. This averaged probability map is then thresholded (in the case of binary segmentation) to generate the final segmentation mask.

This native support not only streamlines the ensemble procedure but also guarantees consistency in preprocessing, and inference behavior across the different models. Furthermore, by relying on the default configuration of **nnU-Net v2**, the ensemble method benefits from the full test-time augmentation (if enabled), patch-based inference, and temperature scaling calibration mechanisms already integrated into the framework.

Thanks to its ease of integration and empirical reliability, the ensemble method was used as a central reference point for comparison with other uncertainty estimation techniques described in this section.

4.3.3. Test-Time Augmentation Implementation

Test-Time Augmentation (TTA) was integrated into the uncertainty estimation pipeline to introduce controlled perturbations to the input data and evaluate the model's predictive stability under slight, realistic variations. As previously discussed in Section 2.7.3, TTA is particularly suited to capture *aleatoric uncertainty*, which stems from inherent noise or ambiguity in the input data rather than from the model's parameters.

In the context of this work, TTA was applied conservatively, with minimal transformations. This choice was driven by the characteristics of the dataset: full-body computed tomography scan (CT) scans, which exhibit very low inter-subject variability in pose and acquisition protocol. Unlike natural images, where objects may appear in a wide range of orientations or scales, clinical full-body MR images are highly standardized—patients are typically scanned in supine position with consistent orientation across acquisitions. Consequently, large geometric perturbations such as 90-degree rotations or heavy elastic deformations would introduce unrealistic configurations, potentially violating the assumption that the augmented inputs remain within the same distribution as the training data.

To remain faithful to the real-world distribution while still inducing useful variability for uncertainty estimation, the following soft augmentations were applied at inference time:

- **Random rotations** around each of the three spatial axes (axial, sagittal, and coronal), with angles uniformly sampled from the interval $[-5^\circ, +5^\circ]$.
- **Random translations** along all three spatial axes, with displacements sampled uniformly in the range $[-5, +5]$ voxels.
- **Intensity scaling**, achieved by multiplying the input volume by a random scalar $\alpha \in [0.9, 1.1]$ to simulate variation in signal intensity across scans.

These transformations were chosen to reflect plausible acquisition-related variations such as patient repositioning, small scanner misalignments, or mild intensity fluctuations due to different hardware or protocols. Moreover, all transformations were designed to be invertible, enabling a precise de-augmentation of the model’s output masks to ensure consistency across predictions.

From an implementation perspective, the TTA mechanism was integrated directly into the prediction pipeline, as illustrated in the code snippet below:

```
if tta == True:
    augment_imgs(original_img, num_aug)
    infer_vols_nnunet_batch(augmented_imgs, ensemble, T)
    take_prob_masks(inferenced_volumes_dir, masks_dir)
    deaugment_masks(transformation_info_path, masks_dir, deaugmented_masks_dir)
    avg_mask_vol = average_confidence_masks_dir(original_mask_path,
        deaugmented_masks_dir)
    final_mask = avg_mask_vol
```

This pipeline proceeds as follows:

1. The original input image is augmented n times using the aforementioned transformations. All transformation parameters (rotation matrices, translation vectors,

scaling factors) are recorded in a structured JSON format for traceability and later inversion.

2. Each augmented volume is processed by the segmentation model (or ensemble) using the standard `nnU-Net v2` inference pipeline, optionally including temperature scaling if specified.
3. The resulting softmax probability masks are extracted and saved.
4. All masks are then *de-augmented*, i.e., warped back into the original reference frame of the input image using the inverse of the recorded transformation parameters.
5. The final mask is computed as the voxel-wise average of the de-augmented probability masks, producing both a final prediction and a dense uncertainty map based on inter-augmentation variability.

The TTA pipeline was designed to be modular and extensible, allowing for easy adjustment of the augmentation intensity or number of samples. However, to balance computational cost with the benefit of improved uncertainty estimation, the number of augmentations was kept between 5 and 10 in most experiments. This range was found to be sufficient to capture stable patterns of variation without introducing significant runtime overhead.

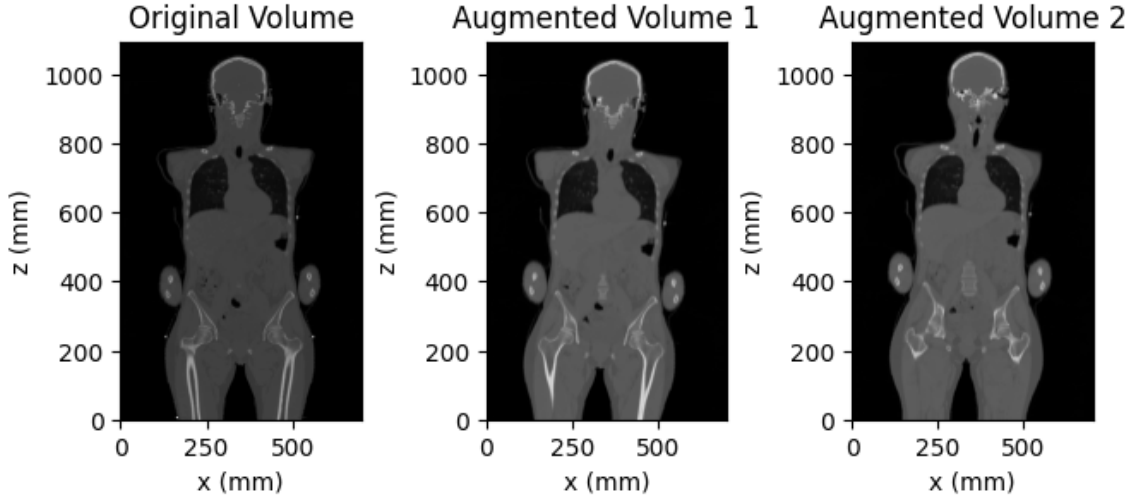


Figure 4.6: Visualization of input volumes used in the Test-Time Augmentation (TTA) process. The image on the left shows the original coronal slice of a full-body CT scan. The two images on the right show the same coronal plane after the application of random soft augmentations, specifically small rotations and translations.

The final output of the TTA process is a probabilistic segmentation mask representing the

average prediction across all augmented views, along with associated uncertainty maps (e.g., voxel-wise variance or entropy). These outputs can then be used for further analysis or visualization.

In summary, the implementation of Test-Time Augmentation in this work was guided by domain-specific constraints and the principle of minimally invasive perturbation. By focusing on realistic transformations and limiting the augmentation budget, the method enables effective estimation of aleatoric uncertainty while preserving the integrity of the input data distribution and maintaining compatibility with the clinical nature of full-body CT.

4.3.4. Checkpoints Ensemble Inference

Once the 30 weight checkpoints were obtained through cyclical training as described in Section 4.2.2, a custom inference pipeline was developed to extract epistemic uncertainty from this ensemble of internal models. Unlike standard **nnU-Net v2** ensemble inference—designed for combining predictions from different folds—this method required sequential inference using the same architecture and training fold but with distinct model snapshots sampled at different stages of training.

To this end, a dedicated Python script was implemented to manage the inference process across all 30 checkpoints. The script accepts as input a single test image and iteratively loads each of the 30 saved model weights. For each checkpoint:

- the model is initialized with the specific saved weights;
- the image undergoes inference using the **nnU-Net v2** framework;
- the resulting softmax probability map is saved for later aggregation.

After collecting the 30 probability maps for a given image, a voxel-wise average is computed across all predictions to obtain the final output mask. Formally, this operation corresponds to:

$$\bar{p}(y^*|x^*) = \frac{1}{n} \sum_{i=1}^n p(y^*|x^*, w_{t_i}),$$

where $n = 30$, and each w_{t_i} represents a checkpointed weight configuration sampled from the training process. The resulting probability map reflects the average predictive belief over the sampled parameter space and thus encodes the *epistemic uncertainty* associated with the model’s parameters.

The inference pipeline was implemented to be modular and scalable, supporting both batch and single-image processing. All intermediate predictions were optionally retained to enable post-hoc uncertainty analysis, such as the computation of voxel-wise prediction variance or entropy.

In practice, this approach allowed for uncertainty quantification without requiring multiple training runs or architecture replications. Although it incurs a higher inference cost compared to using a single model, it offers a computationally efficient alternative to traditional ensembles, since all checkpoints originate from a single training trajectory.

4.3.5. Entropy Mask and Heatmap for Uncertainty Visualization

The uncertainty estimation techniques described in the previous subsections ultimately produce a single voxel-wise probability map as output. These maps reflect the model’s confidence regarding class membership at each spatial location and are typically used for computing binary or multiclass segmentation masks via thresholding (e.g., selecting voxels with probability $p > 0.5$ for the positive class). However, while highly informative for segmentation purposes, these raw probability maps are not directly interpretable as explicit measures of voxel-level uncertainty. For this reason, additional post-processing is required to transform probabilistic outputs into a format suitable for qualitative assessment of uncertainty.

A well-established approach in the literature, also adopted in this work, involves the use of entropy-based transformations to derive uncertainty heatmaps from the predicted probability masks [3, 14]. The rationale behind this method lies in the fact that the entropy of a Bernoulli distribution (appropriate in binary segmentation tasks) provides a compact and intuitive measure of uncertainty: it is maximized when the predicted probability is 0.5 (maximum indecision), and minimized when the predicted probability approaches 0 or 1 (high model confidence).

Formally, for a voxel with predicted probability $p \in [0, 1]$, the associated entropy is defined as:

$$H(p) = -p \log_2(p + \varepsilon) - (1 - p) \log_2(1 - p + \varepsilon),$$

where ε is a small constant (e.g., 10^{-10}) used for numerical stability near the boundaries. This function is symmetric and bounded between 0 and 1, where 1 corresponds to

maximum uncertainty and 0 to full confidence.

To operationalize this concept, a voxel-wise transformation of the softmax output is performed to compute the entropy map. The result is a spatially resolved scalar field where each voxel carries a value encoding the local uncertainty. These values can be conveniently visualized using heatmaps, with warmer colors highlighting more uncertain regions, and cooler colors indicating higher model confidence.

The practical implementation of this transformation in our experiments was done through a custom Python function, which also included optional normalization and customizable thresholding steps to enhance interpretability.

The entropy is computed voxel-wise and optionally normalized within the range $[0, 1]$, with values below a certain threshold set to zero to suppress noise or low-information regions. The result is an entropy map that serves as a proxy for epistemic or aleatoric uncertainty, depending on the source of the underlying probability map (e.g., deep ensemble, test-time augmentation, or checkpoint-based averaging).

The entropy-based heatmaps proved to be an effective visualization tool throughout the experimental phase, enabling the identification of regions where the model was less confident in its predictions. This, in turn, supported a deeper analysis of model behavior, particularly in areas near class boundaries, in ambiguous anatomical zones, or in cases affected by noise or acquisition artifacts. An example of its usage can be visualized in Figure 4.7.

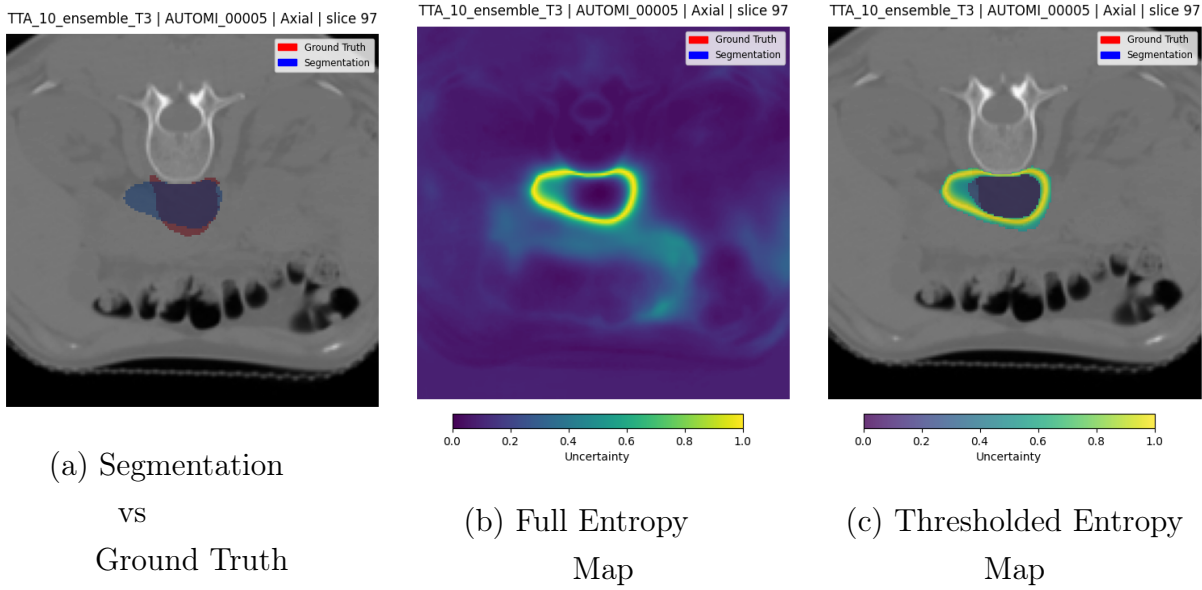


Figure 4.7: Visualization of the same axial slice (slice 97) from subject `AUTOMI_00005`, obtained using the ensemble at which is applied test-time augmentation with 10 augmentation, and temperature scaling $T = 3$. **(a)** shows the predicted binary segmentation (blue) overlaid on the CT scan alongside the ground truth (red). The segmentation was derived by thresholding the softmax probability map at 0.5. **(b)** displays the full entropy map computed from the averaged softmax probabilities, illustrating voxel-wise uncertainty across the region. High-uncertainty areas are highlighted in yellow/green, corresponding to regions where the model entropy is higher. **(c)** shows the thresholded entropy map, where low-uncertainty values have been removed to suppress background noise and enhance interpretability. The threshold value applied here was empirically selected according to the analysis described in Section 5.3.

5 | Results

This chapter presents the experimental evaluation of the methods implemented for uncertainty estimation described in the previous section. The results are structured to progressively address the performance of the system from multiple perspectives: segmentation accuracy, probabilistic calibration, spatial uncertainty behavior, and clinical applicability.

We begin by providing an overview of the evaluation methodology and the organization of the results. We then report the **quantitative metrics** obtained on the held-out test set, including Dice Score, Expected Calibration Error (ECE), and Brier Score. These metrics allow us to evaluate both the segmentation quality and the reliability of the predicted confidence scores.

Next, we turn to a **qualitative analysis**, showing entropy-based heatmaps overlaid on anatomical slices to visually assess how uncertainty is distributed across different anatomical regions. This is followed by an in-depth examination of the **localization power of uncertainty**, with a particular focus on how well high-entropy regions align with prediction errors. We also propose a method to derive **optimal entropy thresholds** that filter out uninformative areas and highlight clinically relevant discrepancies, specially for those methods using temperature scaling calibration with high temperature parameter ($T=3$).

Finally, we present a second evaluation phase involving a new dataset of 8 patients whose segmentation masks were **manually corrected by physicians**. This phase aims to assess whether the uncertainty maps produced by the model coincide with the regions that were actually modified by experts. This provides a realistic assessment of the potential role of uncertainty estimation in supporting human-in-the-loop medical image segmentation workflows.

Overall, this section aims to validate the proposed methods through a multi-faceted evaluation pipeline, ultimately demonstrating the value of uncertainty-aware segmentation in real-world clinical applications.

5.1. Quantitative Results: Segmentation and Calibration Performance

All quantitative results reported in this section were computed on the held-out test set, composed of 5 patients never seen during training. The evaluation focused on three key metrics: Dice Score (mean across all cases), Expected Calibration Error (ECE), and Brier Score. Table 5.1 summarizes the performance of the tested configurations, sorted by decreasing Dice Score. For clarity, the top 3 values in each metric are highlighted in **bold**, while the bottom 3 are colored in **red**.

Table 5.1: Detailed quantitative comparison of uncertainty estimation methods on the 5-case test set. Each cell reports *mean $\pm$ std* with minimum and maximum in parentheses. Bold indicates top-3 results; **red** indicates bottom-3.

Experiment	Dice Score	ECE	Brier Score
TTA_5_ensemble_T3	0.8145 $\pm$ 0.0359 (0.7881–0.8769)	0.0157 $\pm$ 0.0023 (0.0123–0.0187)	0.0375 $\pm$ 0.0064 (0.0283–0.0452)
TTA_5_ensemble	0.8144 $\pm$ 0.0355 (0.7867–0.8755)	0.0305 $\pm$ 0.0079 (0.0195–0.0411)	0.0409 $\pm$ 0.0073 (0.0306–0.0503)
ensemble_T_1_5	0.8140 $\pm$ 0.0364 (0.7897–0.8774)	0.0413 $\pm$ 0.0088 (0.0290–0.0528)	0.0447 $\pm$ 0.0086 (0.0327–0.0559)
ensemble_T2	0.8140 $\pm$ 0.0364 (0.7897–0.8774)	0.0413 $\pm$ 0.0088 (0.0290–0.0528)	0.0447 $\pm$ 0.0086 (0.0327–0.0559)
ensemble_T3	0.8140 $\pm$ 0.0364 (0.7897–0.8774)	0.0413 $\pm$ 0.0088 (0.0290–0.0528)	0.0447 $\pm$ 0.0086 (0.0327–0.0559)
ensemble	0.8140 $\pm$ 0.0364 (0.7897–0.8774)	0.0413 $\pm$ 0.0088 (0.0290–0.0528)	0.0447 $\pm$ 0.0086 (0.0327–0.0559)
TTA_10_ensemble_T3	0.8138 $\pm$ 0.0360 (0.7857–0.8758)	0.0158 $\pm$ 0.0025 (0.0125–0.0181)	0.0373 $\pm$ 0.0061 (0.0287–0.0448)
TTA_5	0.8127 $\pm$ 0.0400 (0.7796–0.8693)	0.0332 $\pm$ 0.0074 (0.0211–0.0371)	0.0412 $\pm$ 0.0059 (0.0323–0.0449)
checkMMens_T3	0.8117 $\pm$ 0.0383 (0.7794–0.8755)	0.0251 $\pm$ 0.0048 (0.0201–0.0331)	0.0378 $\pm$ 0.0059 (0.0291–0.0454)
checkMMens	0.8117 $\pm$ 0.0384 (0.7791–0.8755)	0.0343 $\pm$ 0.0085 (0.0223–0.0460)	0.0421 $\pm$ 0.0080 (0.0304–0.0526)
checkMMens_T_1_5	0.8117 $\pm$ 0.0384 (0.7792–0.8755)	0.0269 $\pm$ 0.0085 (0.0144–0.0382)	0.0401 $\pm$ 0.0076 (0.0289–0.0499)
TTA_5_T3	0.8100 $\pm$ 0.0344 (0.7809–0.8679)	0.0176 $\pm$ 0.0038 (0.0136–0.0225)	0.0383 $\pm$ 0.0051 (0.0308–0.0447)
baseline_prediction_T2	0.8092 $\pm$ 0.0353 (0.7829–0.8702)	0.0288 $\pm$ 0.0077 (0.0171–0.0378)	0.0417 $\pm$ 0.0069 (0.0315–0.0504)
baseline_prediction_T_0_5	0.8092 $\pm$ 0.0353 (0.7829–0.8702)	0.0478 $\pm$ 0.0077 (0.0365–0.0579)	0.0490 $\pm$ 0.0078 (0.0377–0.0591)
baseline_prediction	0.8092 $\pm$ 0.0353 (0.7829–0.8702)	0.0434 $\pm$ 0.0077 (0.0320–0.0533)	0.0464 $\pm$ 0.0075 (0.0354–0.0562)
baseline_prediction_T_1_5	0.8092 $\pm$ 0.0353 (0.7829–0.8702)	0.0373 $\pm$ 0.0076 (0.0257–0.0468)	0.0439 $\pm$ 0.0072 (0.0332–0.0532)
baseline_prediction_T3	0.8092 $\pm$ 0.0353 (0.7829–0.8702)	0.0204 $\pm$ 0.0022 (0.0171–0.0228)	0.0391 $\pm$ 0.0060 (0.0304–0.0467)
baseline_prediction_T_0_25	0.8092 $\pm$ 0.0353 (0.7829–0.8702)	0.0498 $\pm$ 0.0078 (0.0384–0.0599)	0.0503 $\pm$ 0.0078 (0.0389–0.0604)
TTA_5_T_1_5	0.8088 $\pm$ 0.0346 (0.7797–0.8675)	0.0306 $\pm$ 0.0070 (0.0196–0.0385)	0.0417 $\pm$ 0.0064 (0.0321–0.0495)

From Table 5.1, several key observations can be made:

Experimental Naming Convention. The names of the experiments reported in Table 5.1 follow a systematic nomenclature that reflects the main methodological choices involved in each configuration. For instance, the experiment labeled **TTA_5_ensemble_T3** indicates a combination of multiple strategies: **TTA_5** stands for test-time augmentation with 5 augmented versions of the input image, in addition to the original one (resulting in 6 inference passes per patient); **ensemble** means that predictions were averaged across the 5 folds of the ensemble model; and **T3** denotes that temperature scaling was applied



Figure 5.1: Bar plot comparison of all tested methods based on the three main quantitative metrics: Dice Score (left), Expected Calibration Error (ECE, center), and Brier Score (right). Bars are sorted in ascending or descending order depending on the metric, and the best-performing configuration is highlighted in green in each chart.

to the logits with a temperature parameter $T = 3$. When the `ensemble` component is missing (e.g., `TTA_5_T3`), it means that the inference was carried out using only a single model from the ensemble (typically one fold), rather than averaging predictions across all folds. Conversely, a label such as `ensemble_T3` indicates an ensemble inference with temperature scaling, but without test-time augmentation. The prefix `checkMMens` identifies the checkpoint ensemble approach, in which multiple model checkpoints within the training trajectory of a single fold are used to construct an ensemble. When temperature scaling is applied to this strategy, it is reflected in suffixes such as `_T3` or `_T_1_5` (denoting $T = 1.5$). The baseline predictions, such as `baseline_prediction_T2`, correspond to the inference using a single fold model without ensemble, and possibly with temperature scaling. This naming structure allows a compact yet expressive way to describe each configuration and to trace the contributions of each uncertainty estimation technique to the overall performance.

Best-performing methods. The best-performing configuration across all metrics is `TTA_5_ensemble_T3`, which combines five augmentations with ensemble prediction and temperature scaling at $T = 3$. This setup achieves the highest Dice score (0.8145), the lowest ECE (0.0157), and one of the lowest Brier scores (0.0375), confirming both segmentation accuracy and calibration reliability. The second best by ECE and the best one by Brier is `TTA_10_ensemble_T3`, which shows similar behavior but with 10 augmentations. This suggests that modest increases in the number of augmentations can enhance uncertainty quality without compromising Dice score.

Effect of temperature scaling. Temperature scaling has a substantial positive impact on calibration metrics. This is evident in experiments such as `baseline_prediction_T3`, which achieves significantly better ECE and Brier scores than its unscaled counterpart (`baseline_prediction`), even though both share the same Dice score. Conversely, lower temperatures (e.g., `baseline_prediction_T_0_25`) deteriorate all metrics, placing it among the worst-performing configurations overall. This aligns with the theoretical expectation that $T < 1$ increases prediction sharpness, often resulting in overconfident, poorly calibrated outputs.

Effect of Test-Time Augmentation. TTA improves performance when applied carefully. Comparing `TTA_5` with `baseline_prediction` shows consistent improvements in Dice, ECE, and Brier. Moreover, the additional configuration `TTA_5_T3`, which includes temperature scaling, further boosts both calibration and overlap metrics.

Standard Ensemble and Checkpoint Ensemble. All ensemble variants (with or without temperature scaling) show robust Dice scores, but suffer from comparatively poor calibration (ECE ~ 0.041) and Brier scores (> 0.044). These results suggest that while ensembling helps with stability and segmentation overlap, it does not suffice to ensure well-calibrated confidence estimates. In contrast, the `checkMMens_T3` configuration, based on 30 checkpoints sampled during cyclical training, reaches strong overall performance: Brier score of 0.0378 (top 3), highlighting its potential as a lightweight but effective epistemic uncertainty estimator.

Worst-performing configurations. The lowest-performing setups across all metrics are `baseline_prediction_T_0_25`, `baseline_prediction_T_0_5`, and `baseline_prediction`, all lacking temperature scaling or using suboptimal $T < 1$. Their calibration metrics (ECE and Brier) are among the worst, confirming that the default network outputs are overconfident and poorly calibrated without post-hoc correction.

In conclusion, the results demonstrate that combining test-time augmentation, ensemble inference, and temperature scaling leads to models that are both accurate and reliable in their confidence estimates. Checkpoint ensembles also emerge as promising, especially when computational constraints limit training of multiple independent models. Overall, aggressive or poorly tuned temperature scaling (i.e., $T < 1$) and naive de-augmentation strategies (e.g., zero-order interpolation) should be avoided when aiming for trustworthy uncertainty-aware segmentation in clinical imaging applications.

Figure 5.1 provides a visual comparison of the same results shown in Table 5.1, displaying the ranking of all tested configurations across Dice Score, ECE, and Brier Score. The green bar highlights the best-performing setup, `TTA_5_ensemble_T3`, which clearly leads in all three metrics. The plot confirms the insights discussed earlier: ensemble methods combined with test-time augmentation and temperature scaling provide a consistently strong performance across both segmentation accuracy and calibration quality.

From the ECE and Brier charts, it is evident that configurations lacking temperature scaling tend to accumulate in the higher-error region of the plot, underscoring their poor reliability in uncertainty estimation. Notably, the baseline models with low or no temperature adjustment (`baseline_prediction_T_0_25`, `baseline_prediction_T_0_5`) consistently rank at the bottom of both calibration metrics.

Overall, the bar plots offer a complementary perspective to the numerical results, reinforcing the conclusion that uncertainty-aware techniques must balance accuracy with calibration—something best achieved through carefully designed combinations of TTA,

ensemble averaging, and temperature scaling.

5.2. Qualitative Results: Multi-view Visualization of Predictive Uncertainty

To provide a qualitative assessment of the uncertainty estimation methods, Figure 5.2 displays a comparative visualization of six approaches across three anatomical planes: coronal, sagittal, and axial. Each row corresponds to a different method, while each column pair shows the uncertainty heatmap and corresponding segmentation result, respectively. The views are extracted from a representative patient volume.

It is important to note that, due to the anisotropic voxel spacing of the dataset, the axial view is the most spatially reliable for visual interpretation. Specifically, while the coronal and sagittal views suffer from stretched or compressed voxel spacing along the slice thickness axis, the axial view is composed of approximately isotropic voxels, where each pixel can be considered nearly square. This makes the axial plane the most suitable for evaluating fine anatomical structures and inspecting uncertainty localization.

Baseline Prediction. The first row shows the standard baseline prediction obtained via classical inference on a single fold of the ensemble, without any calibration or aggregation. In this case, the uncertainty maps are dominated by narrow high-entropy bands that contour the predicted segmentation boundaries. This behavior is expected, as class boundaries often coincide with prediction uncertainty. However, the predictive uncertainty fails to extend into regions where the model is actually making structural errors—e.g., misclassified or omitted segments—making the map less informative for identifying incorrect predictions. The model appears confident even in regions where the predicted mask significantly deviates from the ground truth, highlighting the limitations of uncalibrated single-model inference.

Baseline with Temperature Scaling ($T = 3$). The second row illustrates the effect of applying temperature scaling alone (with $T = 3$) to the baseline model. The predicted uncertainty is noticeably more dispersed and highlights not only boundary regions but also internal segments. This suggests improved calibration: the model expresses higher uncertainty where predictions are less reliable, especially in lower-probability inner zones. Although the segmentation accuracy is unchanged, the uncertainty map becomes more useful as a diagnostic tool for identifying problematic regions.

Deep Ensemble. The third row corresponds to the ensemble inference using all five folds of the nnU-Net model. Compared to the baseline, the uncertainty maps display

richer structures and more spatial variability, especially around complex anatomical interfaces. Nevertheless, the uncertainty is still predominantly concentrated near the segmentation contours, and internal inconsistencies are not fully captured, limiting its interpretive power.

Test-Time Augmentation (TTA, $n = 5$). The fourth row demonstrates the effect of applying Test-Time Augmentation with five stochastic transformations. Compared to the ensemble alone, the uncertainty maps here are slightly more extended, particularly in regions sensitive to spatial perturbations, suggesting that this method better captures aleatoric uncertainty. However, like the ensemble, it still struggles to flag major segmentation failures that are not along edges.

Checkpoints Ensemble (CheckMMens). In the fifth row, the Checkpoints Ensemble approach is visualized, where 30 model snapshots from different training cycles are aggregated. This technique shows a unique behavior: the uncertainty heatmaps are smoother and exhibit broader confidence gaps in ambiguous regions, especially in the axial view. This confirms its effectiveness in capturing epistemic uncertainty via posterior sampling.

Best Performing Configuration (TTA + Ensemble + $T = 3$). The final row corresponds to the best-performing configuration identified in our quantitative experiments (see Section 5.1): a combination of ensemble inference, five-fold TTA, and temperature scaling with $T = 3$. This approach generates the most informative and well-distributed uncertainty maps. High-entropy regions correlate closely with visible segmentation errors across all three anatomical planes. In the axial view, where resolution is highest and voxels are isotropic, the predictive uncertainty strongly highlights areas of anatomical ambiguity and imperfect boundary alignment. Notably, uncertainty is elevated not only at segmentation edges but also in partial segmentations and low-confidence regions within the organs.

Overall, this qualitative comparison confirms the importance of combining multiple uncertainty estimation strategies. The best-performing configuration not only achieves superior quantitative metrics but also offers uncertainty maps that better correlate with actual segmentation failures, making it a valuable tool for clinical review and semi-automated correction workflows.



Figure 5.2: Comparison of predictive uncertainty maps and segmentation results across three anatomical planes (coronal, sagittal, axial) for six different methods. Each row corresponds to a different uncertainty estimation strategy. For each method, the left column in each plane shows the input image overlaid with the uncertainty heatmap, while the right column shows the predicted segmentation mask overlaid on the original image. Ground truth is shown in red, predicted segmentation in blue. The axial plane, shown in the final pair of columns, offers the most informative view due to its higher and isotropic pixel resolution.

5.3. Uncertainty Localization and Thresholding for Clinical Guidance

While the quantitative metrics presented earlier—such as Dice Score, Expected Calibration Error (ECE), Brier Score, and Reliability Diagrams—offer valuable insights into model segmentation performance and probabilistic calibration, they are inherently limited in describing the spatial structure and interpretability of the predicted uncertainty maps. Specifically, these scalar metrics cannot inform us *where* the model is uncertain, nor whether such uncertainty actually correlates with regions where the model tends to fail. Yet, this spatial dimension is critical in medical imaging applications, especially when uncertainty estimates are intended to guide human reviewers in correcting automated segmentations.

To address this gap, we conducted a detailed analysis focused on the spatial relationship between uncertainty values and voxel-level classification outcomes. Each voxel in the predicted segmentation was categorized into one of four classes: *true positive (TP)*, *false positive (FP)*, *false negative (FN)*, or *true negative (TN)*, based on its agreement with the ground truth. We then compared the distribution of uncertainty values across these four categories to assess whether the model tends to express higher uncertainty in regions where it makes mistakes—that is, FP and FN voxels.

This analysis was performed on the held-out test set, composed of 5 patients that were never seen during training, ensuring a fully unbiased evaluation. By investigating the average uncertainty per voxel class, as well as the full distributions of uncertainty values, we aimed to answer the following critical question: *do higher uncertainty scores reliably highlight the regions where the model is more likely to err?* A positive answer to this question would reinforce the role of uncertainty maps as clinically actionable tools, enabling focused visual inspection and correction in areas most at risk of segmentation failure.

In the following section, we provide both qualitative visualizations and quantitative summaries that elucidate how uncertainty is distributed across TP, FP, FN, and TN regions, and how this distribution varies across different uncertainty estimation strategies.

Focus area definition. Rather than analyzing the entire image volume—which would include large background areas irrelevant to clinical decision-making—we restricted the analysis to a region within 15 mm of the ground truth segmentation. This constraint reflects a realistic clinical scenario, where a physician would only inspect areas near the target anatomy. Figure 5.3 shows an example of the spatial region used in this analysis:

ground truth in blue, and the surrounding buffer area (up to 15 mm) in red.



Figure 5.3: Visualization of the 15 mm buffer region (in red) surrounding the ground truth segmentation (in blue). Only voxels within this area were included in the uncertainty-by-region analysis.

Uncertainty distribution across prediction classes. Using the predicted probability maps and their associated entropy values, we computed the binary classification outcomes for each voxel and grouped the entropy values according to whether they belonged to TP, FP, FN, or TN categories. The aggregated distributions are visualized as boxplots for each experiment. Figure 5.4 and Figure 5.5 show the results for the `checkMMens_T3` and `TTA_10_ensemble_T3` experiments, respectively.

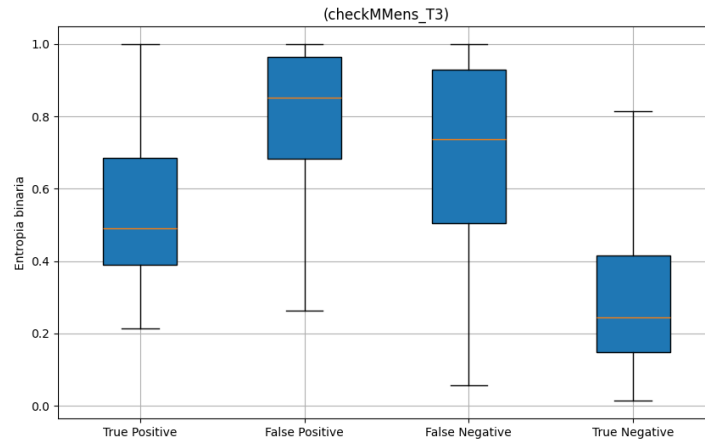


Figure 5.4: Uncertainty distribution by prediction class for the `checkMMens_T3` method. The boxplots reveal that False Positives and False Negatives exhibit significantly higher uncertainty values than True Positives and True Negatives.

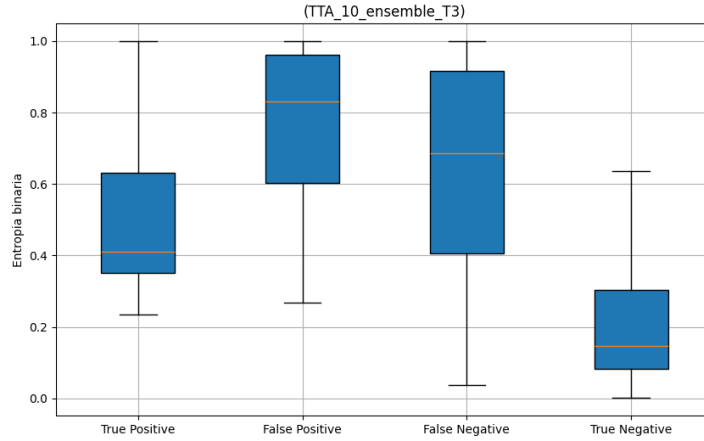


Figure 5.5: Uncertainty distribution by prediction class for the `TTA_10_ensemble_T3` method. A similar trend is observed: incorrectly predicted voxels (FP and FN) show higher entropy values than correctly predicted ones.

These distributions support the hypothesis that uncertainty is meaningfully aligned with segmentation errors: both FP and FN regions exhibit elevated entropy, whereas TP and TN voxels generally have lower uncertainty. This pattern confirms the potential clinical utility of entropy maps as tools for guiding expert review toward problematic regions.

Toward optimal uncertainty thresholding. Based on these findings, we propose a strategy to derive an optimal threshold on the entropy map to filter out low-uncertainty (i.e., high-confidence) regions and retain only those areas that are likely to contain prediction errors. The idea is to suppress the visualization of regions where the model is confidently correct (especially TNs), thereby reducing cognitive load for the reviewing clinician.

To identify this threshold, we defined an objective function that prioritizes the coverage of false predictions while penalizing the inclusion of correct predictions:

$$\text{score}(t) = \text{FP}_{\text{coverage}}(t) + \text{FN}_{\text{coverage}}(t) - [\text{TP}_{\text{coverage}}(t) + \text{TN}_{\text{coverage}}(t)]$$

This function was maximized for each experiment to find the threshold t that most effectively isolates high-risk areas. The coverage functions represent the proportion of voxels in each class with entropy greater than t . The full optimization pipeline was implemented in Python and executed across all experiments and test patients.

To further explore how the entropy landscape changes depending on the calibration strength, we evaluated the optimal threshold for the `checkMMens` method under three

different temperature scaling settings: $T = 1$, $T = 1.5$, and $T = 3$. Figure 5.6 shows the class-specific coverage curves as a function of the entropy threshold. Each curve quantifies how many voxels in a given class exceed a certain entropy value, thus providing insight into how uncertainty is distributed across true and false predictions.

These plots reveal a clear trend: as the temperature T increases, the entropy values across all voxel classes also increase, causing a shift in the optimal threshold toward higher values. This phenomenon is expected, as temperature scaling flattens the softmax distributions, effectively raising the entropy of predictions—even in regions that are confidently correct. In such cases, entropy maps tend to include widespread low-intensity noise across the volume. The thresholding strategy plays a key role here by acting as a filter that attenuates background uncertainty and sharpens the contrast around truly ambiguous regions. This selective suppression of spurious entropy helps emphasize only those areas where the model exhibits substantial doubt. The effect is especially pronounced at $T = 3$, where higher entropy levels would otherwise mask clinically meaningful patterns under a veil of general uncertainty.

In summary, the optimal entropy threshold is not a fixed parameter but a dynamic value that adapts to the characteristics of the calibration method—particularly to the degree of smoothing introduced by temperature scaling. The thresholding approach presented here offers a principled mechanism to clean up the entropy map, reduce visual clutter, and enhance the visibility of potentially erroneous predictions, thus supporting a more targeted and efficient clinical review process.

Robustness of the selected thresholds. In addition to finding the global optimal threshold per experiment (i.e., computed on the full test set), we also computed subject-specific thresholds to assess inter-patient variability. Figure 5.7 shows the comparison for several key experiments. Each bar represents the optimal threshold for a specific patient; the global threshold is shown in a lighter shade of the same color. The results suggest that for most methods, the thresholds remain tightly clustered across subjects, indicating strong robustness and potential for generalization.

Entropy suppression and re-scaling. The usefulness of uncertainty thresholding becomes even more evident in methods that apply temperature scaling with values greater than 1.5. In such cases, the resulting entropy maps tend to be noisier and less interpretable. Applying a threshold in these settings significantly enhances the focus of the uncertainty map by suppressing irrelevant high-frequency noise and amplifying clinically meaningful uncertainty.

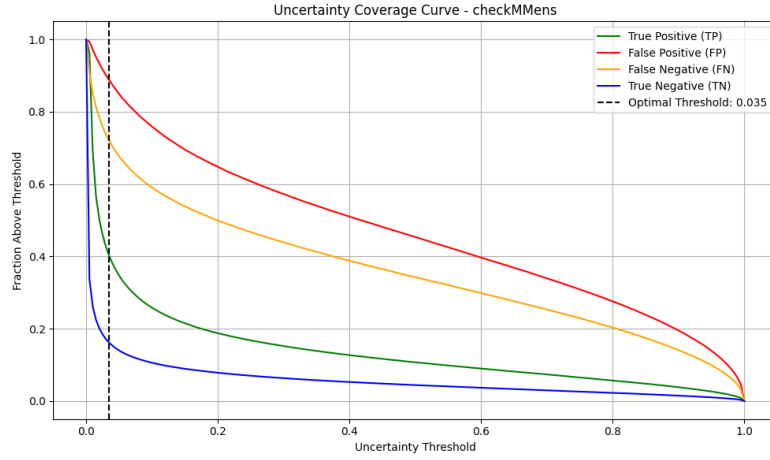
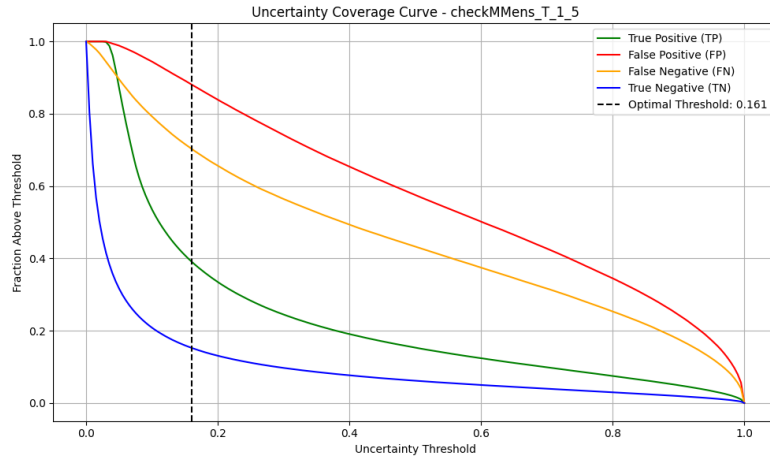
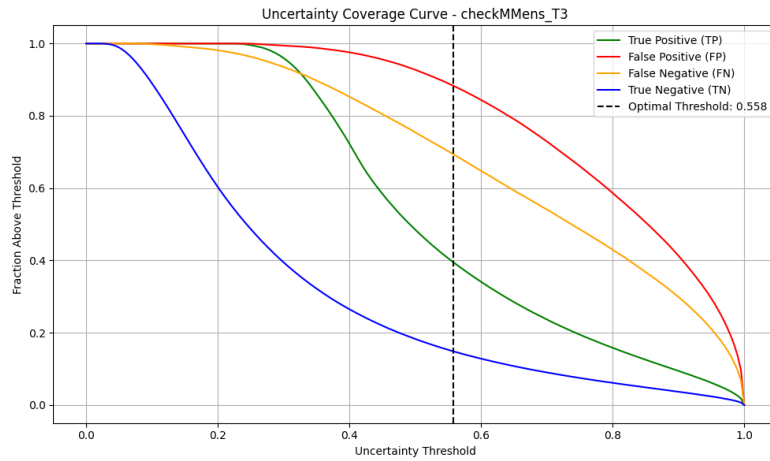
(a) $T = 1$ (b) $T = 1.5$ (c) $T = 3$

Figure 5.6: Coverage curves for the **checkMMens** method with increasing temperature values. Each plot shows the class-wise voxel coverage curves as a function of entropy threshold t . The optimal threshold t^* , which maximizes the score function, shifts to higher values as temperature increases.

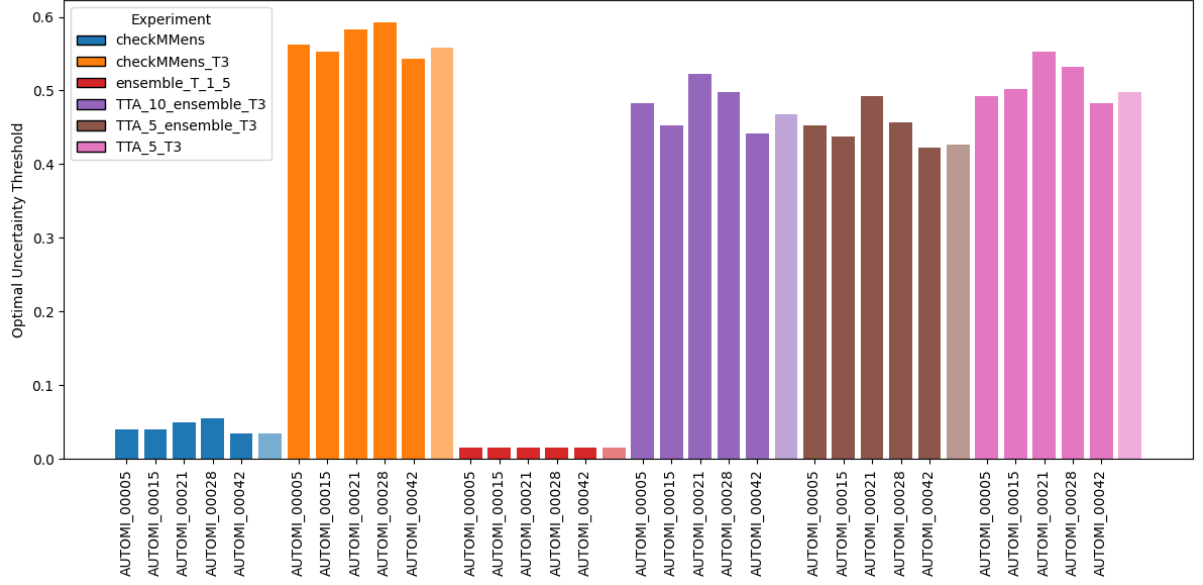


Figure 5.7: Comparison between global and per-subject optimal uncertainty thresholds across selected experiments. Each experiment is color-coded; bars of the same shade but with higher transparency represent the global threshold for that method. Most experiments exhibit low threshold variance across patients, supporting the general applicability of the chosen value.

Once the optimal threshold t^* is determined, we propose to apply a normalization strategy on the entropy map to improve visual interpretability. Specifically, all entropy values below t^* are set to zero (i.e., considered confidently correct), and all values above are re-scaled to the $[0, 1]$ range using:

$$\tilde{H}(x) = \frac{H(x) - t^*}{\max(H) - t^*}, \quad \text{for } H(x) > t^*$$

This procedure enhances the contrast of the most uncertain regions, making them more salient to the reviewer.

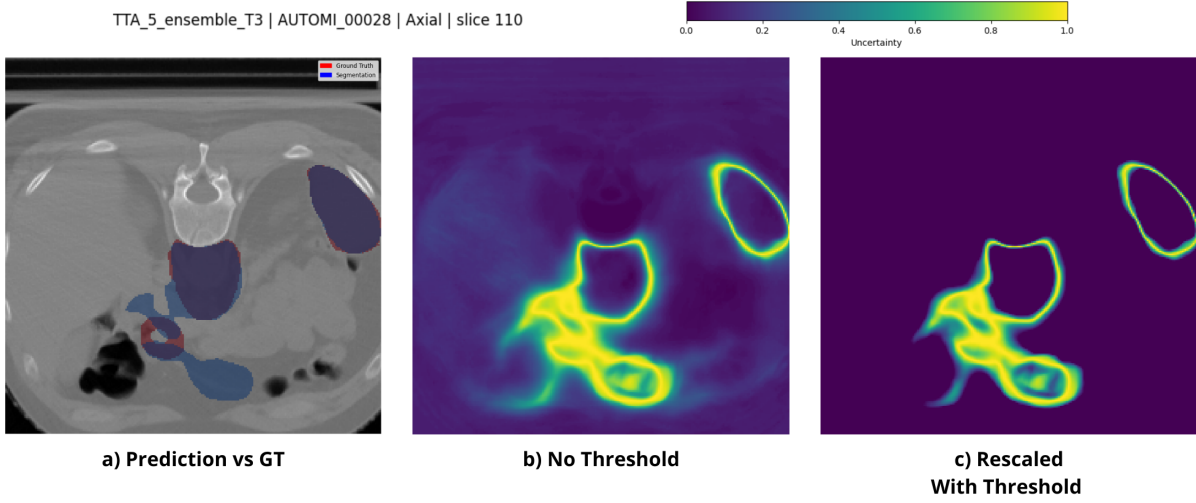


Figure 5.8: Original entropy map (left) vs. thresholded and rescaled version (right) for the TTA_5_ensemble_T3 experiment. Applying the best threshold $t^* = 0.4$ suppresses noisy low-entropy regions and amplifies critical uncertain zones.

This analysis demonstrates that entropy-based uncertainty maps not only correlate with prediction errors but can also be effectively thresholded to isolate regions likely to require manual review. The methodology proposed here offers a principled, data-driven approach to filter uncertainty visualizations, paving the way for real-world integration of uncertainty-aware AI systems in clinical workflows.

Importantly, it should be emphasized that the threshold identified through our optimization procedure is not intended to be a rigid value to be universally applied. Rather, it serves as a well-informed *suggestion*—a recommended starting point tailored to the specific method used. As shown in Figure 5.7, the optimal threshold varies slightly across patients, reflecting anatomical variability, imaging characteristics, and segmentation difficulty. This thresholding approach proves to be especially beneficial for methods employing temperature scaling with high values of T , where the entropy maps tend to be globally elevated and more prone to diffuse background noise. In these settings, applying the optimized threshold becomes essential to isolate only the most uncertain—and potentially erroneous—regions, ensuring a more focused and clinically meaningful visualization.

From a practical standpoint, the proposed threshold can guide users (e.g., radiologists or other clinicians) in selecting an appropriate level of uncertainty suppression. They may interactively adjust the threshold upward or downward based on the context. Increasing the threshold, as our results suggest, tends to further restrict the visualization to only those regions with the highest entropy—i.e., the areas most likely associated with segmentation errors. This behavior supports a more focused and efficient review process,

allowing experts to prioritize inspection of potentially problematic zones while filtering out confidently correct predictions that likely require no attention.

In summary, the entropy threshold is best interpreted as a tunable control parameter rather than a fixed cutoff. Its optimal value—data-derived and method-specific—provides a strong default, but the final decision can and should be adjusted according to clinical priorities, user preferences, and domain knowledge.

5.4. Second Phase Evaluation: Correction-Based Ground Truth

After the extensive evaluation conducted on the initial 5-patient test set—characterized by fully manual ground truth annotations—an additional experimental phase was carried out to further investigate the clinical relevance of uncertainty estimation. This second phase was designed to simulate a more realistic clinical scenario, in which automatic segmentations are first generated by the model and subsequently reviewed and corrected by medical professionals.

Entropy distribution on corrected slices. Before analyzing the results qualitatively, we also conducted an analytical investigation—mirroring the approach used in the first phase—on the distribution of entropy values across voxel-level classification outcomes (TP, FP, FN, TN). As in the previous analysis, we restricted our attention to voxels within 15 mm of the ground truth segmentation to reduce background bias. Importantly, only those slices that were actually revised by clinical experts were included in this analysis, as they are the only ones that provide reliable information on segmentation correctness.

The baseline model used to generate the predictions shown to clinicians was the standard **ensemble** configuration. We first analyzed the uncertainty distribution for this model and then applied different uncertainty estimation techniques to the same volumes, focusing particularly on the best-performing configuration from the first phase: Test-Time Augmentation with 5 augmentations, ensemble inference, and temperature scaling.

The resulting boxplots are shown in Figure 5.9. In each case, the distributions confirm that misclassified voxels (i.e., false positives and false negatives) tend to be associated with significantly higher entropy values than correctly predicted ones (TP and TN). Moreover, we observe that the application of temperature scaling not only increases overall entropy levels, but also compresses the entropy of erroneous predictions into a narrower and more clearly defined range. For example, in the `TTA_5_ensemble_T3` case, the entropy of FP and FN voxels is almost entirely concentrated between 0.7 and 1.0, whereas the entropy of correct predictions remains largely below this range. This clearer separation may improve the interpretability and usability of the uncertainty maps.

Quantitative metrics on corrected regions. To complement the entropy-based analysis, we also computed the main quantitative metrics—Dice Score, Expected Calibration Error (ECE), and Brier Score—on the same corrected slices. Table 5.2 summarizes the results.

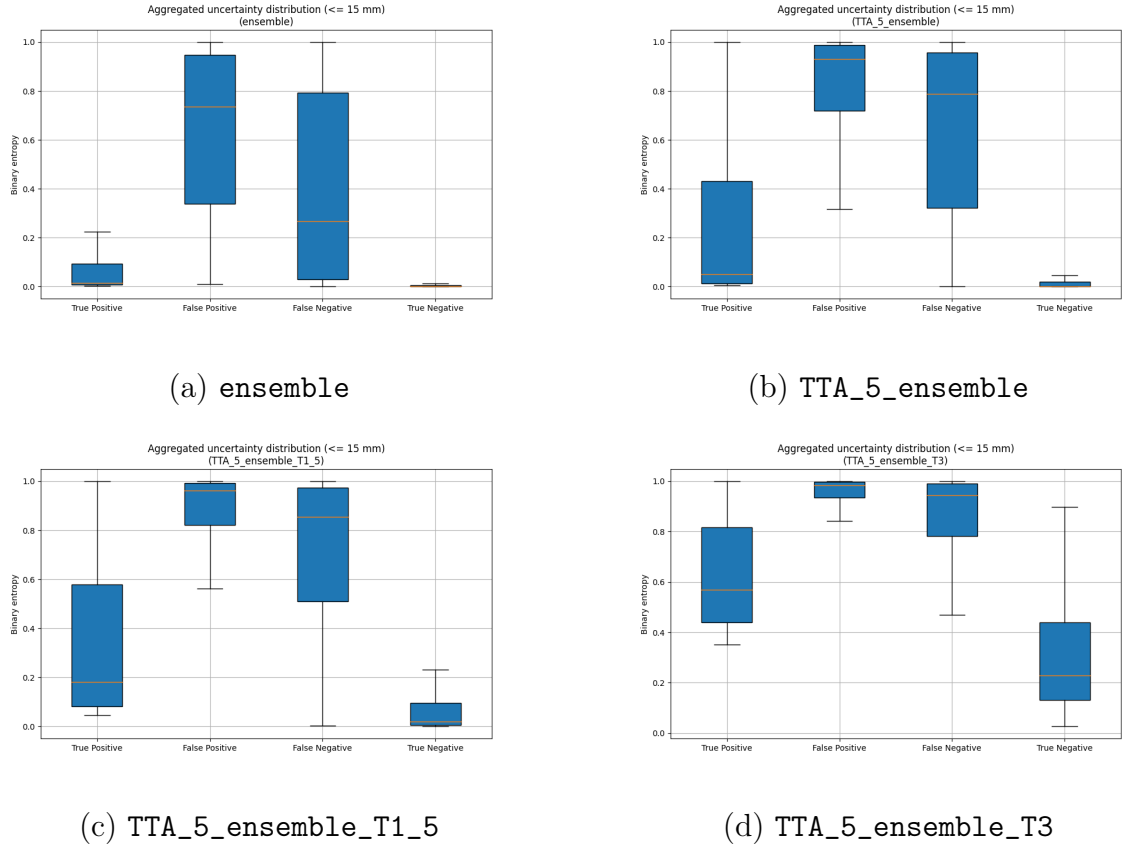


Figure 5.9: Aggregated uncertainty distribution (binary entropy) for four uncertainty estimation methods, computed on slices that were manually corrected by clinical experts. Only voxels within 15 mm of the ground truth segmentation were considered. False Positive and False Negative regions consistently show higher entropy than True Positive and True Negative areas. Temperature scaling increases entropy globally but concentrates it more tightly around erroneous voxels.

Table 5.2: Detailed metrics on manually corrected slices (second evaluation phase). Each value is reported as *mean* $\pm$ *std*. Bold indicates top-3 performance.

Method	Dice Score	ECE	Brier Score
TTA_5_ensemble	0.9114 $\pm$ 0.0264	0.6688 $\pm$ 0.0492	0.00098 $\pm$ 0.00047
TTA_5_ensemble_T1.5	0.9125 $\pm$ 0.0262	0.6678 $\pm$ 0.0490	0.00099 $\pm$ 0.00042
TTA_5_ensemble_T3	0.9107 $\pm$ 0.0235	0.6576 $\pm$ 0.0451	0.00140 $\pm$ 0.00040
ensemble	0.9367 $\pm$ 0.0324	0.6759 $\pm$ 0.0469	0.00079 $\pm$ 0.00057

Although the **ensemble** configuration achieves the best Dice and Brier scores—likely due to the fact that the corrections were performed starting from its predictions—slight improvements in ECE are observed with temperature scaling. This suggests that the modified softmax distributions provide a more honest representation of the model’s confidence, potentially yielding more useful uncertainty maps for downstream clinical review.

That said, these quantitative results should be interpreted with caution, as they are inherently biased by the correction process itself. Nonetheless, they provide additional insight into the trade-offs between accuracy and uncertainty interpretability.

After the extensive evaluation conducted on the initial 5-patient test set—characterized by fully manual ground truth annotations—an additional experimental phase was carried out to further investigate the clinical relevance of uncertainty estimation. This second phase was designed to simulate a more realistic clinical scenario, in which automatic segmentations are first generated by the model and subsequently reviewed and corrected by medical professionals.

Qualitative alignment with expert corrections. To complement the previous analytical findings, we conducted a qualitative evaluation aimed at visually assessing whether regions of high predictive uncertainty coincide with anatomical areas corrected by clinical experts. Here we compare multiple uncertainty estimation methods side-by-side, allowing a more comprehensive interpretation of their potential clinical utility.

Figure 5.10 illustrates the comparison on two axial slices from two different patients that were manually revised by clinicians. For each slice, the rows correspond to different uncertainty estimation methods: **ensemble**, **TTA_5_ensemble**, **TTA_5_ensemble_T1.5**, and **TTA_5_ensemble_T3**. The columns show three views: the segmentation overlaid with the expert-corrected ground truth (left), the full entropy map derived from the softmax probabilities (center), and the rescaled entropy map after applying the optimal threshold (right) discussed in section 5.3.

The visual comparison demonstrates that high-entropy regions often coincide with model errors subsequently corrected by clinicians. Notably, the **TTA_5_ensemble_T3** configuration produces more widespread but better localized uncertainty maps, highlighting both under- and over-segmented regions. In contrast, the **ensemble** baseline produces narrower uncertainty contours, often limited to the immediate borders of the prediction.

The thresholded versions of the entropy maps are especially informative: they remove background noise and reveal the most ambiguous regions—those that overlap with false positives or false negatives in the original prediction. This alignment supports the hy-

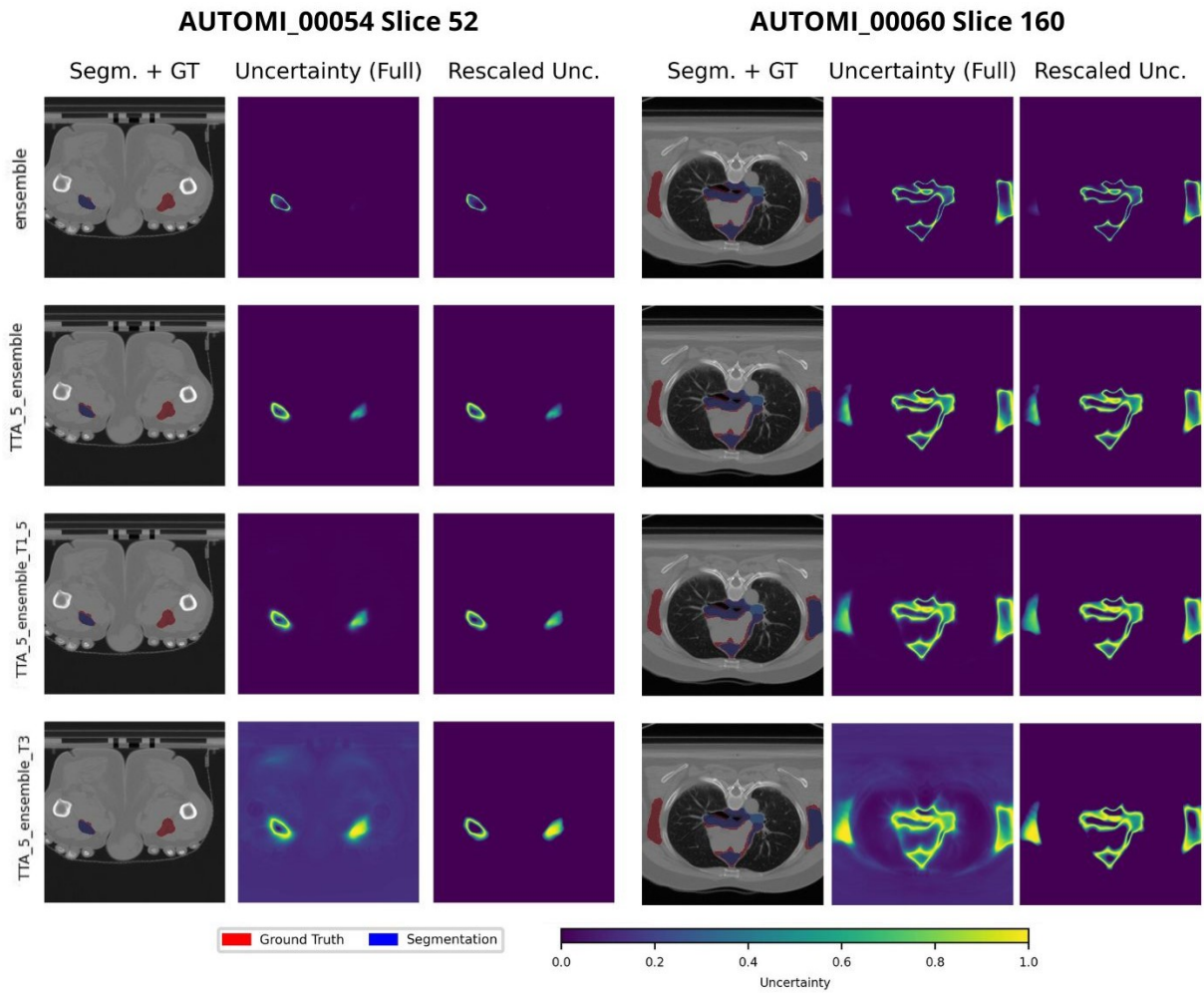


Figure 5.10: Qualitative comparison of different uncertainty estimation methods on two axial slices from patients AUTOMI_00054 (left) and AUTOMI_00060 (right), both revised by clinical experts. Each row corresponds to a different method. Columns show: segmentation overlaid with ground truth (left), full entropy map (center), and rescaled thresholded entropy map (right). Red outlines represent the expert-corrected segmentation, while blue indicates the model prediction. High-uncertainty regions (yellow-green) tend to align with the areas that were corrected, such as boundary mismatches and partial segmentations, particularly in the TTA + Temperature Scaling configurations.

pothesis that uncertainty maps can effectively guide expert review, focusing attention on clinically relevant errors and reducing unnecessary inspection of confidently correct areas.

These qualitative results further reinforce the findings from the quantitative analysis, highlighting the value of combining ensemble methods, test-time augmentation, and temperature scaling to produce interpretable and actionable uncertainty estimates in clinical segmentation tasks.

5.5. Interactive Visualization App: Slice-by-Slice Uncertainty Inspection

In order to facilitate the exploration of the results obtained through uncertainty-aware segmentation, a dedicated interactive application was developed using the Python library `streamlit`. The purpose of this app is to allow researchers and clinicians to visualize, slice by slice, the segmentation outputs and uncertainty maps overlaid on medical images (e.g., CT scans), providing a powerful tool for qualitative inspection and user-in-the-loop analysis.

The graphical user interface (GUI) of the app is organized around several interactive widgets that guide the user through the visualization process. Figure 5.11 shows an example of the interface, where different options are available for customizing the view.

The first dropdown menu, labeled **Experiment**, enables the selection of a specific uncertainty estimation method among those tested (e.g., TTA+Ensemble+ $T = 3$). Once the method is selected, the user can choose a specific patient case via the **Image ID** combobox. In the example shown, results for patient AUTOMI_00060 are being inspected.

The **View Plane** checkbox allows the user to toggle between axial, coronal, and sagittal planes; in this example, the axial view was chosen. Additional checkboxes enable or disable the overlay of the **Uncertainty Map**, **Ground Truth**, and **Segmentation** mask, each rendered with a distinct and easily interpretable color palette: segmentation in blue, ground truth in red, and uncertainty using a viridis colormap.

A key feature of the app is the **Slice Index** slider, which allows scrolling through the individual slices of the selected plane, offering fine-grained control over the inspection process. To limit the field of view and focus on the region of interest, a **Crop Size** parameter can be specified. For instance, if set to 500 mm, the app automatically extracts a square region of size 500×500 mm centered around the predicted segmentation in the selected slice.

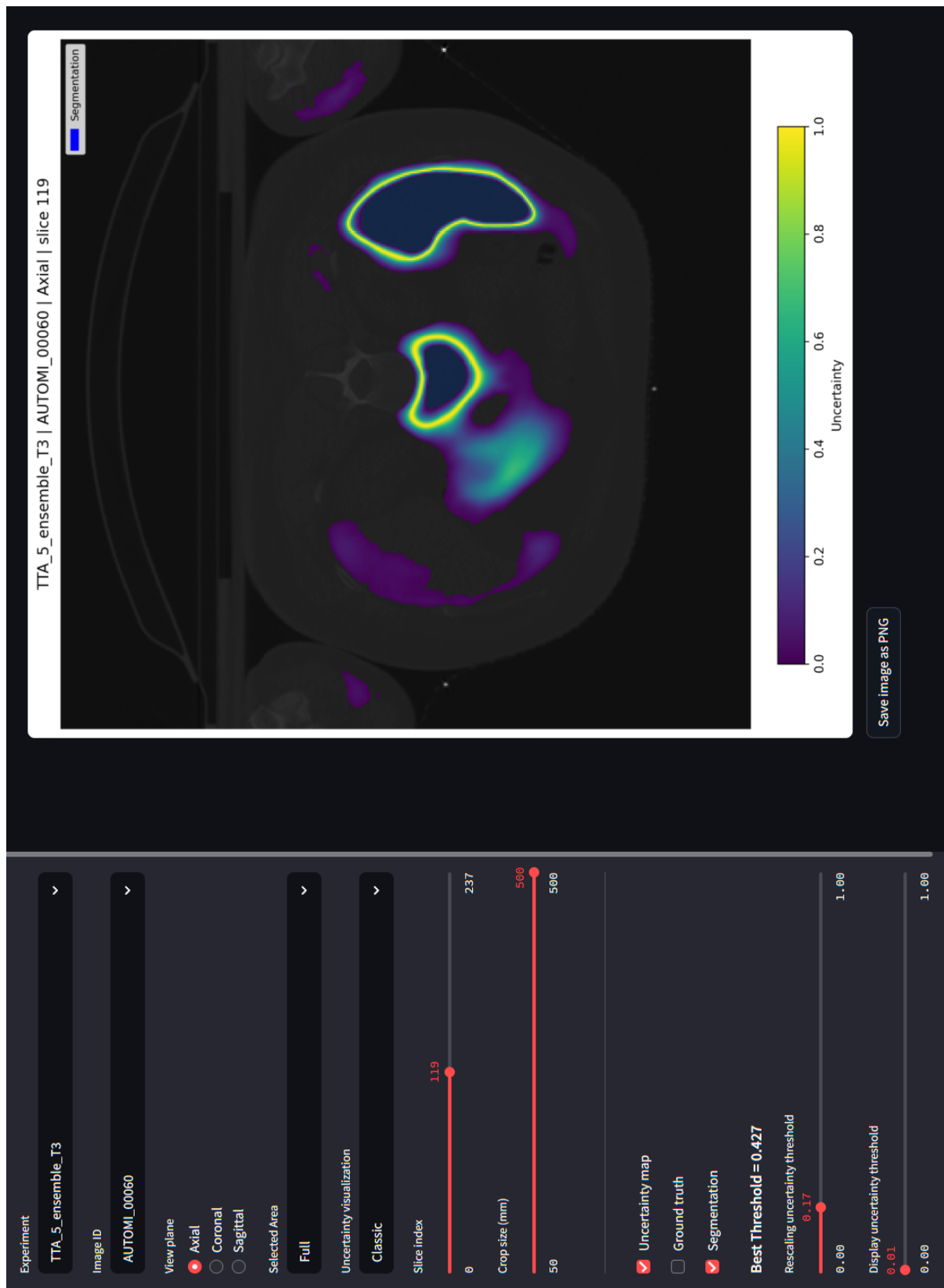


Figure 5.11: Graphical interface of the developed Streamlit app for slice-wise interactive visualization of uncertainty-aware segmentations.

To support interpretability, the interface also displays the **Best Threshold** associated with the selected uncertainty estimation method—corresponding to the optimized entropy cutoff discussed in Section 5.3. Two additional sliders, **Rescaling Uncertainty Threshold** and **Display Uncertainty Threshold**, provide further control over how uncertainty maps are visualized. The first threshold defines the entropy cutoff used for suppressing low-confidence values; all entropy values below this threshold are set to zero, while those above are rescaled to the $[0, 1]$ range. The second threshold then defines which regions of the rescaled map are displayed on screen—i.e., values below this second threshold are rendered transparent.

Another noteworthy control is the **Selected Area** combobox, which determines the spatial region from which uncertainty values should be extracted and visualized. The available options are: **Full**, showing the uncertainty across the entire slice; **Inside Segmentation**, restricting visualization to the predicted organ region; and **Outside Segmentation**, focusing on background or surrounding tissue. This feature provides anatomical context to the uncertainty analysis, enabling finer clinical insights.

To further enhance interpretability, the app also includes the **Uncertainty Visualization** mode selector. Among the options, the **No Borders** mode is particularly relevant, as it suppresses uncertainty around the edges of the predicted mask—often trivially high due to class boundaries—and focuses attention on less expected areas. In this mode, an additional slider labeled **Border effect distance (mm)** appears, allowing the user to control the exponential attenuation of uncertainty within a specified margin (e.g., 3 mm) around the segmentation contours. This helps mitigate the common bias that uncertainty is always concentrated near edges, potentially highlighting internal errors or overlooked regions. Figure 5.12 illustrates this effect: voxels lying exactly on the segmentation contour have their uncertainty set to zero, while nearby regions are smoothly attenuated, enhancing the visibility of internal uncertainty hotspots that would otherwise be hidden by dominant contour effects.

In addition to the previously described visualization options, the app also integrates an advanced inspection mode called **Gamma-U**. This feature extends the concept of uncertainty visualization by not only considering the magnitude of uncertainty but also incorporating its spatial relationship with respect to the segmentation boundaries, a more in depth explanation can be found into the Appendix A.1. The approach takes inspiration from the gamma index used in radiotherapy for validating dose distributions, adapting its principle to the context of medical image segmentation.

While standard entropy maps often highlight areas of inherent ambiguity, they tend to

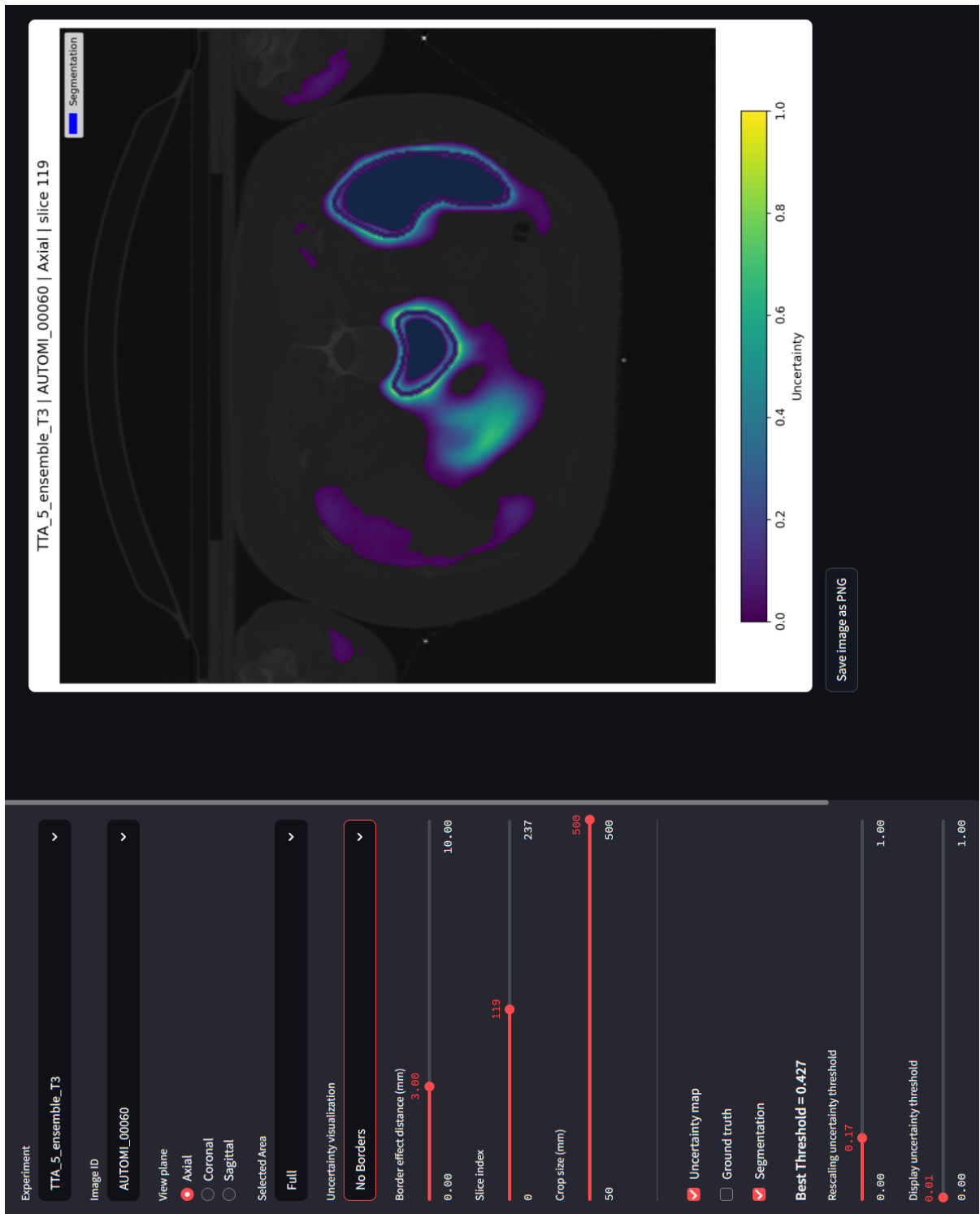


Figure 5.12: Visualization of the uncertainty map using the No Borders mode. Entropy values at the segmentation boundaries are suppressed and rescaled using a 3 mm margin, revealing less trivial uncertainty regions.

overemphasize regions near the contours due to natural label transitions. The Gamma-U visualization overcomes this limitation by combining two complementary aspects: the distance from the segmentation contour and the local uncertainty value. This combination enables a more nuanced identification of regions that might require further scrutiny. Specifically, it highlights areas that are either excessively uncertain despite being far from boundaries or anomalously positioned relative to the expected segmentation shape.

An example of this visualization is shown in Figure 5.13, where the Gamma-U map is applied to an axial slice from patient AUTOMI_00060. The colormap visually encodes the Gamma-U values, with shades of blue indicating well-behaved regions and a transition to red tones for areas where the Gamma-U exceeds the critical threshold of 1. These highlighted zones can correspond to internal misclassifications, ambiguous structures, or peripheral artifacts that the model has not confidently resolved.

As with the other visualization modes, the panel on the left allows the user to interactively adjust key parameters controlling the behavior of the Gamma-U metric. These include the maximum distance considered (**dM**), the maximum uncertainty reference (**UM**), and the relative weights assigned to the spatial and uncertainty components (**alpha** and **beta**). This fine-grained control makes it possible to adapt the visualization to the specific anatomical region or clinical question under analysis, ensuring that the feedback provided is both interpretable and actionable.

The entire interface is designed to respond in real time to user interactions, providing immediate visual feedback as sliders and parameters are adjusted. This dynamic behavior greatly enhances the exploration experience and allows intuitive tuning of visualization parameters.

Overall, the developed Streamlit app represents a powerful and extensible tool for the interactive inspection of segmentation and uncertainty results. It serves both as a research utility—allowing rapid visual validation of experimental results—and as a potential clinical aid, enabling practitioners to identify uncertain regions that may require correction or further examination. Future work may extend this application to include 3D rendering, volumetric navigation, or multi-class segmentation support, further increasing its relevance in real-world clinical workflows.

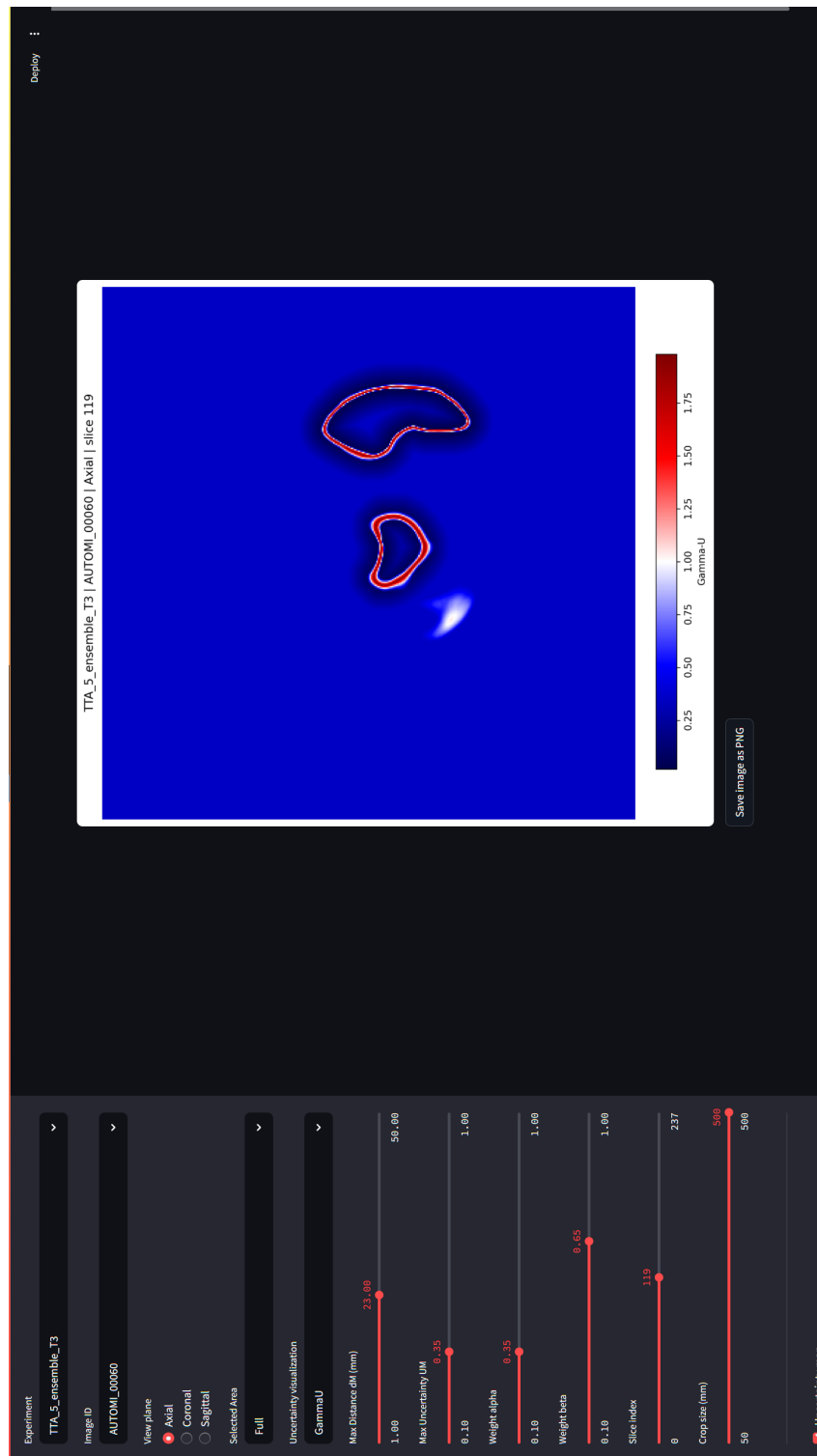


Figure 5.13: Example of the Gamma-U uncertainty visualization mode applied to an axial slice of patient AUTOMI_00060. The color scale reflects the Gamma-U value, with red areas indicating regions where the combined effect of distance and uncertainty exceeds the threshold of 1. This mode enables the identification of potentially problematic regions that are either excessively uncertain or geometrically inconsistent with the segmentation.

6 | Conclusions and future developments

In this thesis, we addressed the problem of predictive uncertainty estimation in the context of medical image segmentation using deep learning, with a particular focus on full-body CT scans of the lymphatic system. Recognizing the inherent ambiguities and clinical risks associated with automated segmentation, especially when applied to complex anatomical structures such as lymph nodes, we explored a variety of techniques designed to quantify and visualize model uncertainty in a meaningful and actionable way.

The core of our experimental framework was based on **nnU-Net v2**, a state-of-the-art, self-configuring segmentation pipeline that enabled robust training even with a limited number of annotated cases. Building on this foundation, we implemented and compared several complementary uncertainty estimation methods, including temperature scaling for calibration, deep ensembles, test-time data augmentation (TTA), and a checkpoint-based ensemble approach using cyclical learning rate schedules. Additionally, we introduced entropy-based heatmaps for qualitative visualization and developed an interactive Streamlit application to enable intuitive, slice-by-slice inspection of results.

Quantitative evaluations on a held-out test set of five patients revealed that the combination of ensemble inference, moderate test-time augmentation, and post-hoc temperature scaling with $T = 3$ yielded the best overall performance across multiple metrics, including Dice score, Expected Calibration Error (ECE), and Brier score. Importantly, we observed that while ensembles provided robust predictions, their confidence estimates remained poorly calibrated unless explicitly corrected. Temperature scaling emerged as an effective and computationally inexpensive solution to this problem, especially when applied after probabilistic aggregation via TTA or ensemble averaging.

Our qualitative analysis further confirmed that uncertainty maps, particularly those based on entropy, were not only theoretically informative but also aligned well with actual prediction errors. By studying the spatial distribution of uncertainty in relation to false positives and false negatives, we demonstrated that uncertainty-aware methods can effec-

tively highlight regions prone to segmentation failure. Moreover, we proposed a principled approach for thresholding and rescaling entropy maps, showing that optimal cutoffs can be learned and generalized across patients to improve clinical interpretability.

Perhaps the most clinically significant finding of this work came from the second evaluation phase, where we analyzed 15 additional patient scans corrected by expert physicians. Entropy maps derived from the proposed uncertainty estimation pipeline consistently identified the regions modified by clinicians—thus validating the practical utility of our methods in a realistic human-in-the-loop segmentation workflow. These results suggest that uncertainty estimation can be a powerful tool for guiding expert review, reducing annotation time, and improving the trustworthiness of AI-assisted diagnosis.

Limitations and Future Directions

Despite the encouraging outcomes, several limitations remain. First, the relatively small size of the dataset (45 annotated volumes, with 8 additional corrected cases) limits the statistical generalization of the results. While **nnU-Net** was explicitly designed for small data regimes, future work should aim to evaluate these methods on larger and more diverse datasets, including multi-institutional scans and different anatomical targets. This would allow for a more robust validation of uncertainty estimation strategies and a more comprehensive understanding of their limitations in clinical deployment.

Second, the methods employed in this study focus predominantly on binary segmentation and single-class outputs. Extensions to multi-class or instance-level segmentation, as well as domain adaptation to other imaging modalities (e.g., CT, PET), remain open and promising directions. In particular, the design of uncertainty estimation strategies that can operate in real-time or low-resource clinical settings is of paramount importance for practical adoption.

From a methodological standpoint, further integration of uncertainty into the training process itself—not just as a post-hoc analysis—represents an exciting avenue for research. For example, uncertainty-aware loss functions or adaptive sampling schemes based on predictive entropy could be explored to directly improve model learning. Another promising direction lies in combining epistemic and aleatoric uncertainty into unified uncertainty-aware optimization strategies.

Additionally, while the entropy-based thresholding procedure introduced in this work proved effective in highlighting prediction errors, it was evaluated as a global strategy, independent of anatomical context. A potentially valuable extension would be to investigate

whether the optimal uncertainty threshold varies across different anatomical regions. For instance, areas that are easier to segment (due to clearer contrast or well-defined structure) might exhibit higher optimal thresholds, while anatomically complex or ambiguous regions might require lower thresholds to adequately flag uncertain predictions. Conducting such an analysis—particularly on a larger and anatomically annotated dataset—could lead to the development of region-aware or anatomy-specific thresholding strategies, further enhancing the clinical precision and interpretability of uncertainty maps.

Finally, while entropy maps were effective for visualization, future work could explore more advanced spatial representations of uncertainty (e.g., topological uncertainty graphs, boundary uncertainty maps) or incorporate interactive feedback mechanisms into the user interface to support semi-automatic refinement and active learning. The integration of uncertainty estimation into broader clinical decision-support systems remains a compelling and necessary challenge for the next generation of medical AI tools.

Final Remarks

In conclusion, this thesis provides a comprehensive exploration of uncertainty estimation in medical image segmentation, spanning theoretical insights, practical implementations, quantitative evaluations, and clinical validation. The proposed methods and tools contribute to the broader vision of making AI systems not only accurate but also interpretable, trustworthy, and truly assistive in the clinical decision-making process.

By enabling models to say not only “what” they predict but also “how sure” they are, we take a crucial step toward safer, more transparent, and more human-centered medical AI.

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A | Appendix A

A.1. Gamma-U Index: Combining Distance and Uncertainty

In radiotherapy, the **gamma index** is a widely used metric for comparing two dose distributions—typically the planned and the delivered one—by assessing their spatial and dosimetric agreement simultaneously. The classical gamma index $\gamma(x)$ for a given spatial location x is defined as:

$$\gamma(x) = \min_{x'} \sqrt{\left(\frac{\|x - x'\|}{\Delta d}\right)^2 + \left(\frac{D(x) - D(x')}{\Delta D}\right)^2}$$

where Δd is the spatial tolerance and ΔD is the dosimetric tolerance. A gamma value $\gamma(x) < 1$ indicates that the test dose distribution lies within acceptable bounds of the reference distribution in both location and intensity. This metric is crucial in quality assurance procedures to ensure treatment safety and accuracy.

Inspired by this concept, we propose an analogous metric—called **Gamma-U**—for uncertainty-aware segmentation in medical imaging. Instead of comparing dose values, the Gamma-U index fuses two key elements:

1. the spatial distance from the predicted segmentation contour, and
2. the local uncertainty predicted by the model (e.g., normalized entropy).

This combined index allows for the visualization of potentially problematic regions where uncertainty is high and the voxel is far from the segmentation boundary, highlighting cases of false negatives or erroneous over-segmentation.

Mathematical Formulation

Given a 2D binary segmentation map `seg2d` and a corresponding uncertainty map `unc2d`, the Gamma-U index $\Gamma_U(r)$ at each pixel r is computed as:

$$\Gamma_U(r) = \sqrt{\alpha^2 \cdot \left(\frac{\min(d(r), d_M)}{d_M} \right)^2 + \beta^2 \cdot \left(\frac{U(r)}{U_M} \right)^2}$$

Where:

- $d(r)$ is the distance from pixel r to the nearest segmentation boundary,
- $U(r)$ is the normalized uncertainty value at pixel r ,
- d_M and U_M are saturation thresholds for distance and uncertainty, respectively,
- α and β are weighting coefficients balancing the two terms.

This formula corresponds directly to the implementation in the Streamlit app:

```
def compute_gamma_u_map(seg2d, unc2d, dr, dc, dM, UM, alpha, beta):
    contour = find_boundaries(seg2d)
    distance_map = distance_transform_edt(~contour, sampling=(dr, dc))
    distance_term = alpha * np.minimum(distance_map, dM) / dM
    uncertainty_term = beta * unc2d / UM
    gamma_map = np.sqrt(distance_term**2 + uncertainty_term**2)
    return gamma_map.astype(np.float32)
```

Pixels with $\Gamma_U(r) > 1$ are considered to fall outside the "tolerance zone" and may deserve special attention during qualitative inspection.

Interpretation and Use Cases

This formulation effectively highlights regions where segmentation errors are more likely to occur and is especially valuable for identifying:

- *Internal false negatives*: areas not segmented but with high internal uncertainty,
- *Peripheral false positives*: segmented regions with high uncertainty far from the contour.

In the Streamlit application, Gamma-U is implemented as one of the visualization modes, fully interactive and controllable via sliders for the parameters d_M , U_M , α , and β . This

configurability allows users to adapt the visualization to different anatomical targets or clinical contexts.

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